



Enabling technologies for visualization and analysis of single cell RNA-Seq data

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July 31, 2022

TECHNISCHE UNIVERSITÄT MÜNCHEN

Wissenschaftszentrum Weihenstephan für Ernährung, Landnutzung und Umwelt

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Vollständiger Abdruck der vom Wissenschaftszentrum Weihenstephan für Ernährung, Landnutzung und Umwelt — Lehrstuhl M12 (Mathematische Modellierung biologischer Systeme) der Technischen Universität München zur Erlangung des akademischen Grades eines

Doktors der Naturwissenschaften (Dr. rer. nat.)
genehmigten Dissertation.

Vorsitzender:

Univ.-Prof. Dr. M. Hrabě de Angelis

Prüfer der Dissertation:

- Univ.-Prof. Dr. Dr. F. J. Theis
- 

Die Dissertation wurde am _____ bei der Technischen Universität München eingereicht und durch die Fakultät Wissenschaftszentrum Weihenstephan für Ernährung, Landnutzung und Umwelt am _____ angenommen.

Danksagung

Der Ort, an dem ich ab 2015 die meiste Zeit verbringen würde war mir klar, als Carsten Marr mich nach meinem Masterpraktikum fragte ob ich in seiner Gruppe an Fabian Theis' Institute of Computational Biology (ICB) promovieren möchte. Ich bin den beiden dankbar für die Möglichkeit, in den folgenden Jahren in der wunderbaren Gesellschaft des ICB, insbesondere ihrer beiden Gruppen in denen ich Mitglied sein durfte. Die Arbeitsatmosphäre, die Retreats und Konferenzen waren mir persönlich und fachlich eine Bereicherung. Ich möchte Carsten außerdem für seine Hilfsbereitschaft und fachliche wie persönliche Unterstützung danken, sowie Fabian für seine Anregungen.

Ich danke meiner Familie und meinen Freunden dafür, dass sie mich in dieser Zeit unterstützt haben, besonders unter meinen Lamentationen bezüglich mangelnden Fortschritts beim Verfassen diverser Arbeiten. Ich bedanke mich insbesondere bei meiner Mutter Eva Vitzthum-Angerer und meiner Großmutter Anna Vitzthum für ihren Glauben an mich und dafür dass sie mir in den heißen Schreibphasen den Rücken freigehalten haben. Ich danke meiner wundervollen Partnerin Desiree, die mich im Endspurt mit unendlicher Liebe unterstützt hat. Last but not least danke ich meinen Freuden die mich immer wieder im Leben geerdet haben: Julian, die Marias, Charly, Tina, Tanzeem, Alvaro, Jade, Lena, Jonas, Hannah, Gannie und all die anderen, sowie all den Freunden die im ICB immer beim Kaffee ein offenes Ohr hatten: die [KC]aros, Maren, Alex, Paul, Lisa, Valerio, Benny, Akshaya und alle anderen.

Abstract

Since its inception 11 years ago, single cell RNA sequencing (scRNA-Seq) allows measurement of high-dimensional transcriptome data with single cell resolution. Their availability and accessibility grants unprecedented opportunities for research on disease mechanics and cellular development. This potential prompted ongoing growth in experimental capabilities, data size, and analysis tools, including a zoo of dimensional reduction methods to preprocess and make sense of high-dimensional data. Due to the resulting combinatorial explosion of options, arriving at a set of best practices involves both a lot of research and implementation effort. Traditionally, this effort resulted in inflexible processing and analysis pipelines of which only few experience continuous updates to the newest best practices. Furthermore, mixing and matching analysis steps of pipelines designed around ad-hoc data formats is not a trivial task, especially with the high volumes of data commonly used today. This development shows a clear need for a central, efficient data structure and an extensible framework for analyzing and visualizing the data it represents.

This work describes advances towards a consolidated and accessible scRNA-Seq analysis and visualization experience using such a framework. It focuses on the development of reusable analysis building blocks and algorithms for dimensional reduction tailored to this kind of high-volume, high-dimensional data. A general scRNA-Seq analysis workflow is explained together with the challenges and opportunities related to each step. My contributions outline the development towards this framework in context of **this workflow**: First, data is collected and **quantified**. This step commonly involves **untargeted scRNA-Seq using a commercial platform, which can get expensive for high volumes of data. BART-Seq is an inexpensive protocol for targeted transcriptome quantification for large amounts of samples. Fatma Uzbas, Micha Drucker and I designed its demultiplexing and quantification strategy together, while I also implemented the quantification pipeline**. Second, **data** is read into a data structure able to hold raw quantities, metadata, and different representations such as dimensional reductions. I designed and implemented **AnnData**, a by now widely used generic Python data structure for annotated matrices such as scRNA-Seq expressions. Third, an integrated framework and compatible tools are used to operate on the data structure and visualize insights. **AnnData** is central to **SCANPY**, an efficient Python framework with building blocks for preprocessing, analysis, and visualization of scRNA-Seq data. I co-developed **SCANPY** with F. Alex Wolf, with me focusing on interaction design and usability. Given the central role of dimensional reduction in analysis of high-dimensional data, I developed two methods for creating and **making sense** of them. I developed and continuously improved **DESTINY**, an efficient implementation of the diffusion map algorithm for dimensional reduction in R, which I optimized **and** extended for biological data. Based on an initial idea by F. Theis, I designed the gene relevance algorithm, which detects genes that drive a given dimensional reduction.

The established projects `ANNData`, `SCANPY`, and `DESTINY` prove to be popular contributions to single cell transcriptome analysis. Their advances in accessibility, performance and reproducibility improve the field, while their use by researchers and industry shows promise to power many applications up to and including clinical ones.

Zusammenfassung



List of contributed articles

The present thesis is cumulative. Its results have been published in peer reviewed journals, and are listed below in chronological order.

- i) Angerer, P., Haghverdi, L., Büttner, M., Theis, F. J., Marr, C., & Buettner, F. (2016). Destiny: diffusion maps for large-scale single-cell data in R. *Bioinformatics*, 32(8), 1241–1243. doi:10.1093/bioinformatics/btv715
- ii) Angerer, P., Simon, L., Tritschler, S., Wolf, F. A., Fischer, D., & Theis, F. J. (2017). Single cells make big data: New challenges and opportunities in transcriptomics. *Current Opinion in Systems Biology*, 4, 85–91. doi:10.1016/j.coisb.2017.07.004
- iii) Wolf, F. A., Angerer, P., & Theis, F. J. (2018). Scanpy: large-scale single-cell gene expression data analysis. *Genome Biology*, 19(1), 15–15. doi:10.1186/s13059-017-1382-0
- iv) Uzbas, F., Opperer, F., Sönmezer, C., Shaposhnikov, D., Sass, S., Krendl, C., Angerer, P., Theis, F. J., Mueller, N. S., & Drukker, M. (2019). BART-Seq: cost-effective massively parallelized targeted sequencing for genomics, transcriptomics, and single-cell analysis. *Genome Biology*, 20(1). doi:10.1186/s13059-019-1748-6
- v) Angerer, P., Fischer, D. S., Theis, F. J., Scialdone, A., & Marr, C. (2020). Automatic identification of relevant genes from low-dimensional embeddings of single-cell RNA-seq data. *Bioinformatics*, 36(15), 4291–4295. doi:10.1093/bioinformatics/btaa198

Contents

1	Introduction	1
1.1	How to Understand Cell Behavior	1
1.2	Workflow for scRNA-Seq Data Processing	4
1.2.1	Library Preparation and Sequencing	4
1.2.2	Counting of transcripts or reads	6
1.2.3	Preprocessing of counts	6
1.2.4	Analysis of cellular dynamics, heterogeneity and gene roles	8
1.2.5	Visualization as Analysis steps and result	10
1.3	Organizing Analysis Workflows	12
1.3.1	Analysis Pipelines: Status Quo	12
1.3.2	Frameworks for modular scRNA-Seq Analysis Pipelines	14
2	Methods	16
2.1	Dimensionality Reduction	16
2.1.1	Spectral Decomposition	17
2.1.1.1	Principal Component Analysis	17
2.1.1.2	Diffusion Maps	18
2.1.2	Learned Embeddings	19
2.1.3	Gene Relevance	20
2.2	ANNData	23
2.3	SCANPY	25
2.3.1	Preprocessing	26
2.3.2	Analysis	26
2.3.3	Visualization	26
2.4	BART-Seq (demultiplexing and quantification)	27
3	Summary of contributed articles	29
3.1	destiny: diffusion maps for large-scale single-cell data in R (Angerer et al., 2016)	29
3.2	Single cells make big data: New challenges and opportunities in transcriptomics (Angerer et al., 2017)	31
3.3	Scanpy: large-scale single-cell gene expression data analysis (Wolf, Angerer, & Theis, 2018)	31
3.4	BART-Seq: cost-effective massively parallelized targeted sequencing for genomics, transcriptomics, and single-cell analysis (Uzbas et al., 2019)	32
3.5	Automatic identification of relevant genes from low-dimensional embeddings of single-cell RNA-seq data (Angerer et al., 2020)	33

4	Conclusion and Outlook	35
4.1	Multiresolution scRNA-Seq Analysis	35
4.2	Multimodal SCANPY and ANNDATA	36
4.3	New Dimensions of SCANPY Scalability	37
A	destiny: diffusion maps for large-scale single-cell data in R (Angerer et al., 2016)	57
B	Single cells make big data: New challenges and opportunities in transcriptomics (Angerer et al., 2017)	61
C	Scanpy: large-scale single-cell gene expression data analysis (Wolf, Angerer, & Theis, 2018)	69
D	BART-Seq: cost-effective massively parallelized targeted sequencing for genomics, transcriptomics, and single-cell analysis (Uzbas et al., 2019)	75
E	Automatic identification of relevant genes from low-dimensional embeddings of single-cell RNA-seq data (Angerer et al., 2020)	93

1 Introduction

1.1 How to Understand Cell Behavior

Every aspect of existence as thinking beings touches cell function. Our body consists of cells performing the most varied tasks in order to maintain it: they constitute organs, muscle, and everything else apart from teeth and bone, which are produced by cells. We are protected by the complex interplay of the cells in our immune system, and the physical barrier of the dead keratinous cells making up our outer skin layer. Even our thinking depends on neurons, our psyche on cellular regulation of our brain chemistry. Formed by imperfect evolution, this machinery is not built to last. Health, life and death are therefore topics every human has to confront at some point in life.

Medicine has performed incredible feats of extending our expected span and quality of life. In our consequently aging society, it becomes increasingly important to understand the long term effects of living. The more diseases we can effectively counter, the more it is possible to shift our focus to rarer or more complex ones. **This includes diseases like cancer, caused by mutated cells, or diseases caused by malfunctioning cell regulation. All those diseases have in common that they cannot be understood without knowing the cellular mechanisms involved.**

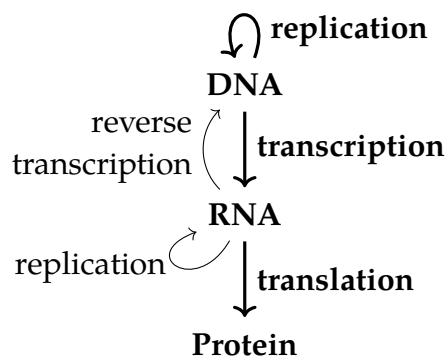


Figure 1.1 The central dogma

Each inner process of a cell is performed directly or indirectly through proteins reshaping themselves while binding to metabolic molecules and each other. Their tasks range from simply transporting oxygen in red blood cells, **over** processing food into readily available energy, to complex developmental **chain reactions** forming our bodies in the womb. Proteins achieve their versatility through the binding characteristics of their twenty different constituent amino acids. Knowing what proteins are being produced in a cell at a given moment **allows to** draw conclusions about its purpose and health. Proteomics, the

measurement of proteins directly, is however made difficult by their complex binding **behavior**. This results in proteomics being limited to targeted analysis of a small number of known proteins (Manzoni et al., 2018). The assembly of proteins in cells is however fairly well understood and measurable. The central dogma of molecular biology states that genes on our chromosomes are transcribed to short lived messenger RNA (mRNA) molecules, which are then translated to proteins (**seed** Figure 1.1). The presence of mRNA in a cell therefore indicates that its respective protein is currently being produced. It is possible to quantify and subsequently analyze snapshots of all mRNA molecules in a cell sample using RNA sequencing (RNA-Seq, Morin et al., 2008). This untargeted large-scale process is called gene expression analysis. During the past decade, biological research worked with averaged RNA profiles of cell samples called bulk RNA-Seq.

Those profiles allow to capture the gene expression of a tissue or fluid sample without accounting for cell-to-cell variability.

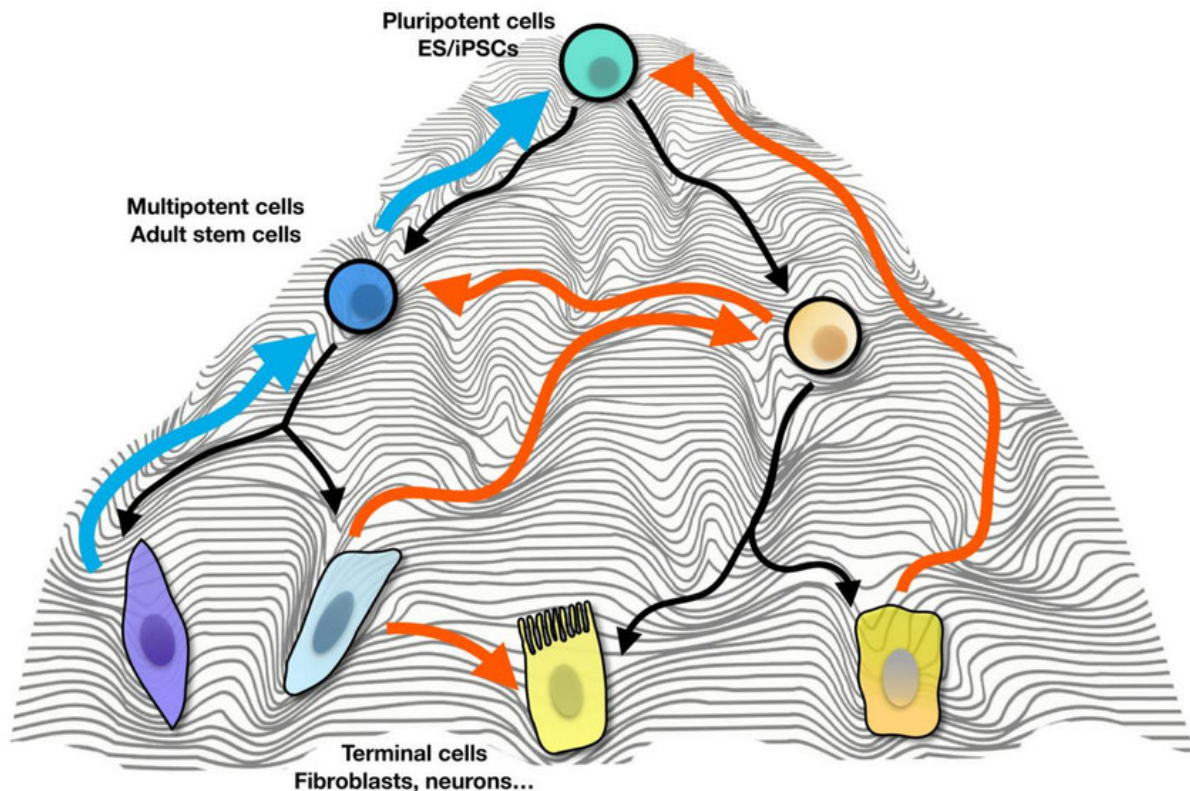


Figure 1.2 Cells differentiate smoothly and in stochastically diverging paths, from high potency (top) to high specialization (bottom). Dedifferentiation (blue arrows) occurs naturally in plants and regenerating basal animals, but not humans. The natural occurrence of transdifferentiation (orange arrows) in humans has only been observed from α to β cells in pancreatic islets. Experiments to artificially induce trans- or dedifferentiation have shown promise for regeneration of tissue damage and effects of aging.

(Figure adapted from Gam, Sung, and Prasad Pandurangan (2019) with permission)

Single cell RNA-Seq (scRNA-Seq) can measure extremely small expression values down to a single mRNA molecule (Tang, Lao, & Surani, 2011). This data comes with its own challenges, but enables the study of heterogeneous tissue, where expression averages from bulk RNA-Seq are not meaningful (Svensson, Vento-Tormo, & Teichmann, 2018). Nature Methods has named scRNA-Seq its method of the year 2013 together with scDNA-Seq (Nature Methods, 2014). The ability to dissect heterogeneity allows to improve our understanding of the complex cellular interaction, in developing, mature, and diseased tissue. This enables new approaches to fight diseases like cancer (Patel et al., 2014; Qian et al., 2017) and influencing cellular development (see **Figure 1.2**). The second great advantage is that the amounts of data present in scRNA-Seq are necessary for machine learning approaches. Advances in microfluidic devices have led to experimental protocols able to sequence more than a million cells (Klein et al., 2015; Macosko et al.,

2015; Zheng et al., 2017). The ample variation between this multitude of individual cells **allows** to derive patterns no human can detect by manual and careful isolation of bulk samples (Huang et al., 2018). For example Plass et al. (2018) have determined the developmental lineage of every cell in a planarian, a regenerative species whose cellular mechanisms could be the key to prevent scarring or delay aging in humans. Vento-Tormo et al. (2018) have profiled cells from human placentas for regulatory interactions in the maternal-fetal interface, furthering our understanding of **pregnancy** diseases. The international community of scientists working on transcriptomics is coming together to create the HUMAN CELL ATLAS (see **Figure 1.3**, Regev et al., 2017).

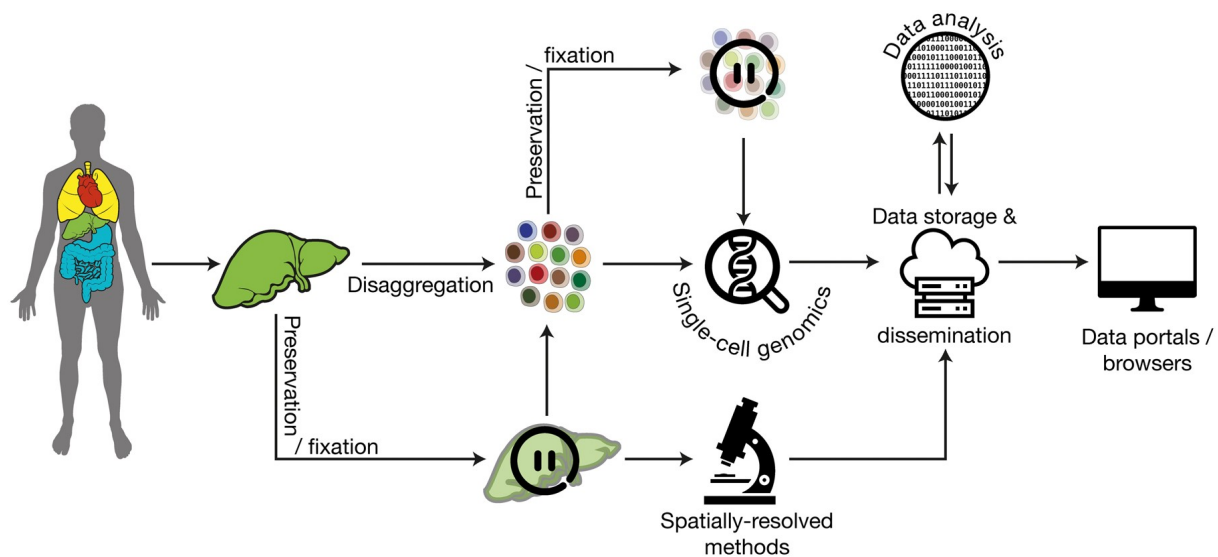


Figure 1.3 Human Cell Atlas (HCA) data flow overview. Next to scRNA-Seq methods operating on disaggregated cells, HCA collects spatial information.

(Figure adapted from Hon et al. (2018) with permission)

All these opportunities come at the cost of additional complexity and more confounding variables (biases). For one, scRNA-Seq has by far fewer mRNA molecules per sample than bulk RNA-Seq, which results in noisier measurements in general (Brennecke et al., 2013). Experimental protocols used for sequencing single cell transcriptomes have different tradeoffs between sensitivity, cost, time investment, and individual biases (Vieth et al., 2019; Ziegenhain et al., 2017). For example, microfluidic protocols cannot completely avoid lysing and sequencing two cells in the same droplet (doublets). The number of cells that can be measured at once with a certain quality is therefore practically limited. Measuring thousands or more cells without exploding costs means that shallow sequencing depths and additional biases are being tackled using computational solutions where applicable (Angerer et al., 2017). For example there are dedicated methods for doublet filtering (McGinnis, Murrow, & Gartner, 2019; Wolock, Lopez, & Klein, 2019). Not all effects however are correctly understood and **approached**. **So have studies** suggested in past years that microfluidic protocols inflate the number of zeros in the data due to technical reasons, where a zero stands for a gene not being detected in a cell at all. Svensson (2020) however asserts that the high number of zeros stems from biological variation.

Increased complexity in analysis can be conquered with improved modeling or more sophisticated preprocessing. From-scratch modeling is more correct, but less flexible, as it requires careful informed design across the whole analysis process. Each new analysis that significantly deviates from previous analyses requires a re-thinking and re-building of the model almost from the ground up. There are however also modeling approaches that instead of performing analysis, replace several preprocessing steps specifically for scRNA-Seq data, like the normalizing single cell dimensionality reduction methods scVI (Lopez et al., 2018) and GLM-PCA (Townes et al., 2019).

The more practical approach is to start with careful preprocessing, transforming data in a way that fulfills the assumptions of subsequent analysis (Vieth et al., 2019). The second step is then to create and use streamlined building blocks for pipelines to preprocess and analyze the data, and to then visualize the insights gained. Until few years ago, there were no freely available programming environments that allowed this approach for single cell data analysis (Stuart et al., 2019; Wolf, Angerer, & Theis, 2018). Analysis pipelines came custom-built for a certain publication, just as hard to take apart and reuse as a holistic model would have been. Every graduate student and postdoctoral researcher had to start from scratch. The thesis at hand is a slice of the author's continued efforts to change this.

In the following **Section 1.2**, I will provide an overview of a general scRNA-Seq analysis workflow. Afterwards, **Section 1.3** will address the organization and reuse of analysis workflows like it. The extent of detail in describing the individual steps will correspond to the extent of my contributions, a detailed breakdown of which follows in **Section ??**.

1.2 Workflow for scRNA-Seq Data Processing

The general workflow of scRNA-Seq data processing consists of five steps. The first step, library preparation and sequencing, is experimental and is performed in a biological laboratory. The second step, counting, creates a cell $\times$ gene matrix from raw data produced in step one. The remaining steps are preprocessing, analysis and visualization of this matrix.

1.2.1 Library Preparation and Sequencing

The first step begins with an experiment to prepare and sequence a transcript library containing mostly mature mRNA but also unspliced mRNA and other RNA (see **Figure 1.4a**). There are different kinds of library preparation protocols, all of which have in common that they isolate and fragment the RNA per single cell, reverse transcribe it into complementary DNA (cDNA) and then tag each fragment with a cell specific barcode (see **Figure 1.4b**). Due to the low amount of RNA in single cells and the protocols' detection limits, all current protocols also contain a step amplifying the available cDNA (Islam et al., 2014). The most important differences between protocols manifest in

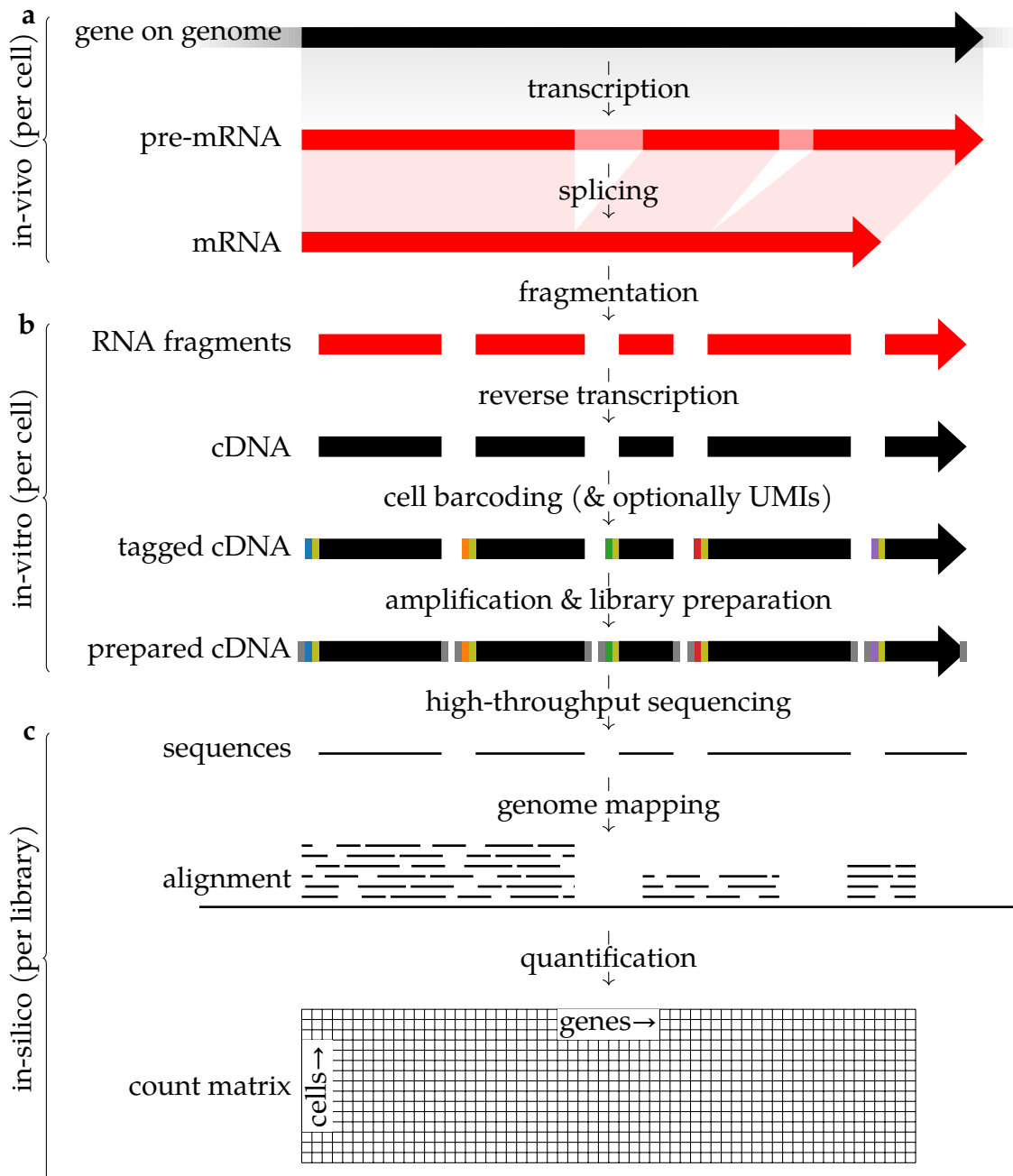


Figure 1.4 ScRNA-Seq library preparation, sequencing, and mapping. **a)** In-vivo steps performed by the cell. A gene is transcribed from DNA to a pre-messenger RNA (pre-mRNA). Introns are spliced out to form one of several possible splicing variant mRNAs. **b)** In-vitro steps performed using lab equipment. The mRNA strands are fragmented chemically and reverse transcribed into cDNA for stability, as RNA degrades quickly. The fragments are tagged with a cell-specific barcode and optionally molecular and library identifiers. The tagged fragments are pooled, amplified, then sequenced using a high-throughput sequencing machine. **c)** In-silico steps performed using a PC or server. The tagged fragment sequences are aligned to a genome or transcriptome. The mapped fragments are counted per cell-gene combination, accounting for splicing variants.

cost, sensitivity and the distinction between those capturing paired fragments starting at both ends of the RNA and those capturing only the RNA end where translation into proteins starts, called the 3' end (Ziegenhain et al., 2018). The sequencing machine finally digitizes the fragments and allows all further processing to happen on a computer (see **Figure 1.4c**). Additional to the cell barcode, some protocols allow to enrich the library with information that can be used to reduce sources of noise or bias. This includes e. g. unique molecular identifiers (UMIs), which **allow to count each transcript only once**, despite the amplification step (Islam et al., 2014). A second example is adding known quantities of RNA from a foreign organism (spike-in RNA) to each cell library, which later **allows** to correct for nonlinear per-cell biases in transcript counts (Lun et al., 2017).

1.2.2 Counting of transcripts or reads

Counting starts from the **paired sequences** read from both ends of a fragment (paired-end reads) or the 3' end sequences (single-end reads). The read or read pair includes a cell barcode and possibly a UMI barcode. Classic counting uses specialized short read alignment tools like BOWTIE (Langmead & Salzberg, 2012; Langmead et al., 2009) or STAR (Dobin et al., 2013) to match reads to a reference genome annotated with information about where genes start, end, and can be spliced (see **Figure 1.4c**). Aligned reads are counted per splicing configuration that translates to a protein, while collapsing the counts for reads with identical UMIs – if available (see **Figure 1.4c**). Since only the resulting count matrix is needed for many kinds of downstream analysis, there have been advances in pseudo-alignment algorithms like KALLISTO (Ntranos et al., 2016) or SALMON (Patro et al., 2017), which efficiently count without a separate alignment step. For ease of use, pipelines wrapping those tools for certain experimental protocols exist. Those include CELL RANGER for the 10X GENOMICS CHROMIUM protocol (Zheng et al., 2017) or zUMIs (Parekh et al., 2018) for several protocols involving UMIs.

Protocols like the above produce counts for all expressed genes, resulting in a sparse matrix of not very sensitively captured expression values. The present thesis details a targeted RNA-Seq protocol called BART-Seq (Uzbas et al., 2019) (see **Section 2.4**). Its tradeoffs are the inverse, allowing to sensitively capture expression values of a low number of target genes for a high number of samples.

1.2.3 Preprocessing of counts

To obtain comparable quantities, counts have to be preprocessed to correct for various technical biases. **Commonly one starts by eliminating rows and columns from the count matrix that are unlikely to be (living) cells or important genes. This filtering removes rows and columns containing a low count sum or very few entries.** Rows matching these criteria are assumed to be empty droplets or wells, while matching columns are genes which due to their low counts do not impact later analysis steps. **It also removes cells containing a high fraction of mitochondrial gene counts, indicative of dead or dying cells**

(Ilicic et al., 2016). Doublets cannot be detected reliably by a simple count threshold, as a pair of small cells can have similar numbers of expressed genes and counts as single large cell, but more sophisticated solutions are emerging (McGinnis, Murrow, & Gartner, 2019; Wolock, Lopez, & Klein, 2019). Luecken and Theis (2019) point out that each of **those filtering covariates** in isolation is **not sufficient** to identify a non-cell row, and suggests conservative joint scoring using more than one covariate.

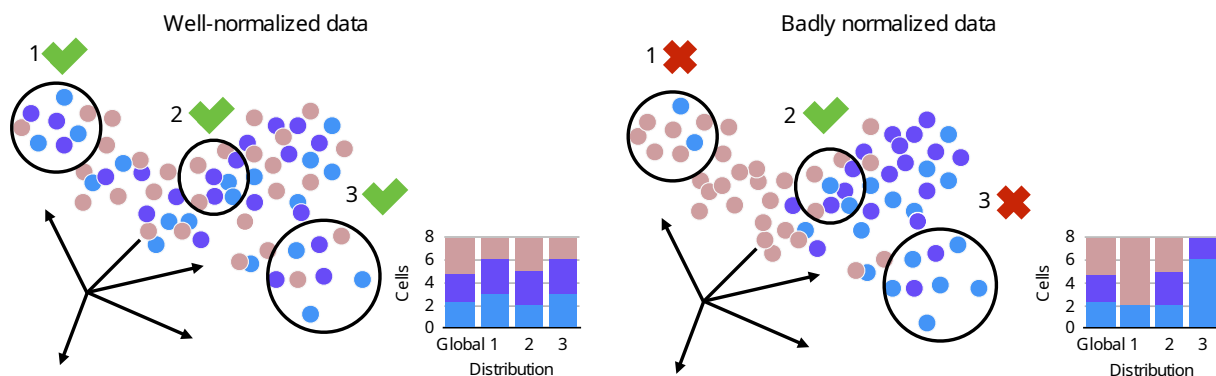


Figure 1.5 Batch effect correction reduces biases between data gathered in multiple sequencer runs. Cells from well-normalized data will have similar expression profiles to cells from the other sequencer runs.

(Figure reproduced from Büttner et al. (2019) with friendly permission)

Another common preprocessing step is normalization. It entails correction for common technical sources of cell variation, as well as biological sources with a strong effect, such as the cell cycle: when not studying the cell cycle itself, its variation can obscure the processes under investigation (Büttner et al., 2015). If multiple runs of the sequencer were used to obtain the full data, there will be biases per run that can be reduced by using a dedicated batch effect correction algorithm (see **Figure 1.5**). While older bulk correction effects such as ComBat (Johnson, Li, & Rabinovic, 2007) have proven to be effective in the last years, a growing number of dedicated scRNA-Seq batch effect correction tools is becoming available (Haghverdi et al., 2018; Korsunsky et al., 2019; Lotfollahi, Wolf, & Theis, 2018; Polański et al., 2020). Preprocessing can also involve scaling count values using a per-cell function derived from the aforementioned spike-ins. All transformation have to be carefully balanced with their potential to over-correct and remove relevant biological variation.

A final common preprocessing step highly variable gene selection. It reduces data by discarding genes with low variability, as they contribute little to the data's variability using differently rigorous algorithms (Brennecke et al., 2013; Yip, Sham, & Wang, 2018). This more complex filtering is performed in addition to the aforementioned quality-based gene filtering, reducing the retained number of genes to the low thousands (Luecken & Theis, 2019).

One step lies between preprocessing and analysis. In order to model and analyze count data, it is almost always necessary to calculate distances between cells' expression profiles. Distance measures usually expect data with no mean–variance dependency, i. e. normally distributed data, which can be fulfilled in two ways. One is to transform

expressions into a space where those expectations hold, commonly by logarithmizing it. The intuition is that in log-space, distances represent log-fold changes, **an** usual way to compare expression levels (Brennecke et al., 2013; Eling et al., 2018). Another way is to employ dimensional reduction methods to embed the data into a lower-dimensional space with the desired properties, such as Principal Component Analysis (PCA, Abdi & Williams, 2010) or diffusion maps (Coifman et al., 2005). An advantage of operating on a low-dimensional embedding instead of on the expression level **are** the reduced computational and memory costs. **However, Svensson (2020) suggested that log-transforming data is a possible cause for distorting data by zero-inflation.** With single cell data specific transforms like SCTransform (Hafemeister & Satija, 2019), as well as embeddings like scVI (Lopez et al., 2018) or GLM-PCA (Townes et al., 2019), promising alternatives are emerging, despite them losing the log-fold interpretability.

1.2.4 Analysis of cellular dynamics, heterogeneity and gene roles

Analysts work interactively, with gradual insight gain shaping further steps. To this end, overview statistics and visualizations are employed from the earliest to the last steps of an analysis (see **Section 1.2.5**).

Analysis of single cell data generally focuses on uncovering previously unknown heterogeneity and its sources (Angerer et al., 2017; Patel et al., 2014; Poirion et al., 2016). The kind of heterogeneity to uncover depends on the kind of data **available**: the simplest case is discrimination between discrete subpopulations of **few clearly different** cell types. A common goal there is to find a small number of marker genes able to reliably distinguish a subpopulation that is sufficiently distinct from the rest of the cells. Methods performing this search are called differential gene expression (DE) analyses (Soneson & Robinson, 2018). While DE analysis methods stem from bulk RNA-Seq, single cell specific methods such as SCDE (Kharchenko, Silberstein, & Scadden, 2014), MAST (Finak et al., 2015), and BASiCS (Vallejos, Richardson, & Marioni, 2016) outperform bulk methods due to the additional information in single cell data.

Another source of heterogeneity is the composition of the studied tissue. Tumors as well as healthy tissue are composed of cells fulfilling a multitude of different roles that can only be uncovered using single cell analysis (Patel et al., 2014; Qian et al., 2017; Zhang et al., 2016). A common method to make sense of **complex heterogeneous data** is cluster analysis. For single cell data, the multi-resolution community detection algorithm LEIDEN (Traag, Waltman, & van Eck J., 2019) and its predecessor LOUVAIN **is** used most commonly, despite Kiselev, Andrews, and Hemberg (2019) arguing **its** lack of performance for smaller datasets. **A general problem of clustering is data reuse.** Clusters are commonly the basis for further analysis like inter-cluster DE analysis – a task the aforementioned single cell specific DE tools are specifically designed for. However, this leads to both clusters and markers being derived from the same data, and treating clusters as ground truth in this setting leads to inflated confidence in the found markers. The problem can be circumvented by joint clustering and DE analysis,

manually propagating uncertainties (Zhang, Kamath, & Tse, 2019), or not relying on clustering to detect genes (Angerer et al., 2020).

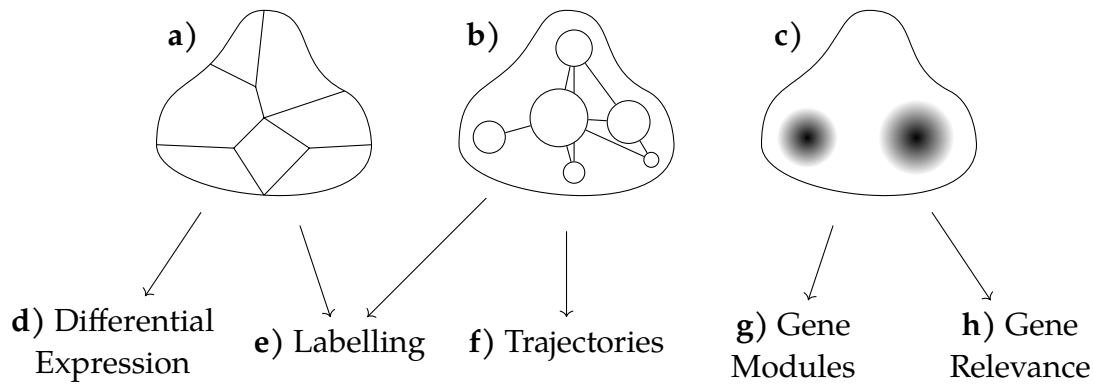


Figure 1.6 Heterogeneity Analysis approaches. For visualization purposes, cells are represented in a dimensionally reduced space represented by the outer border of **a**, **b**, and **c**. All three represent common starting points for further analysis. **a)** Clustering divides the set of cells into discrete subsets (clusters) based on similarity. The clarity of division between clusters shows how easily a given pair of clusters can be discriminated between. **b)** Transition State Analysis dissects metastable states and their transitions. This can accurately represent cell development or more gradual cell identities than discrete types. **c)** Scoring results in each cell having an individual score in one or more schemas, defining likelihood or strength of a certain property. **d)** Differential Expression Analysis allows to identify genes defining a given subdivision, marking them for further analysis and providing insight into cell identities. **e)** Labelling of clusters assigns them a cell type label, based on marker genes and other external knowledge. **f)** Trajectory Analysis groups and orders cells by similar transition state rather than absolute similarities and therefore allows to reason about continuous processes like cellular development or spatial gradients. **g)** Gene Module Analysis integrates previous knowledge about gene roles to gain insights about the likelihood of each cell fulfilling a given role. **h)** Gene Relevance Analysis relates the location of a cell in the embedding back to the genes driving that localization.



While cluster analysis strongly benefits from the availability of scRNA-Seq data, the field of trajectory analysis can only exist because of it. Traditionally, cells were thought of exhibiting a discrete cell type each. For cells that are dramatically specialized for one main role, this idea still holds. Most tissues however contain a multitude of cells fulfilling multiple roles each, without fitting into a discrete and homogeneous type. This includes cells in the process of differentiating from a malleable pluripotent state to a more or less specialized mature state (Haghverdi, Buettner, & Theis, 2015). While marker genes for

certain cell roles or developmental steps exist, this complex and gradual nature of most tissues requires sophisticated analysis and visualization of single cell data. **An approach that uncovers a developmental direction and hierarchy in developmental snapshot data is called trajectory analysis.** Early analysis tools like WANDERLUST (Bendall et al., 2014), or DPT (Haghverdi et al., 2016) can be effectively applied to simple developmental data. **But while cells usually get increasingly specialized during their path of development, the reverse direction (dedifferentiation) can also be observed (see Figure 1.2).** Furthermore, developing cells do not necessarily follow all steps of differentiation in the same order or magnitude. This results in cells temporarily or sometimes permanently stuck as intermediate types. The field of trajectory analysis is therefore constantly evolving to enable the analysis of complex tissues up to whole animals (Plass et al., 2018) using tools like SPRING (Weinreb, Wolock, & Klein, 2018) or PAGA (Wolf et al., 2019).

Analyses depend on the availability of informative data. Since classic scRNA-Seq is limited to spliced mRNAs, additional data modalities are necessary to infer dynamics like directionality in trajectory inference (Efremova & Teichmann, 2020). **As an example, RNA velocity indicates the direction in which cells differentiate and therefore enhances trajectory inference with directionality and more accuracy. It is reconstructed from the relation of spliced and unspliced mRNA** (La Manno et al., 2018).

1.2.5 Visualization as Analysis steps and result

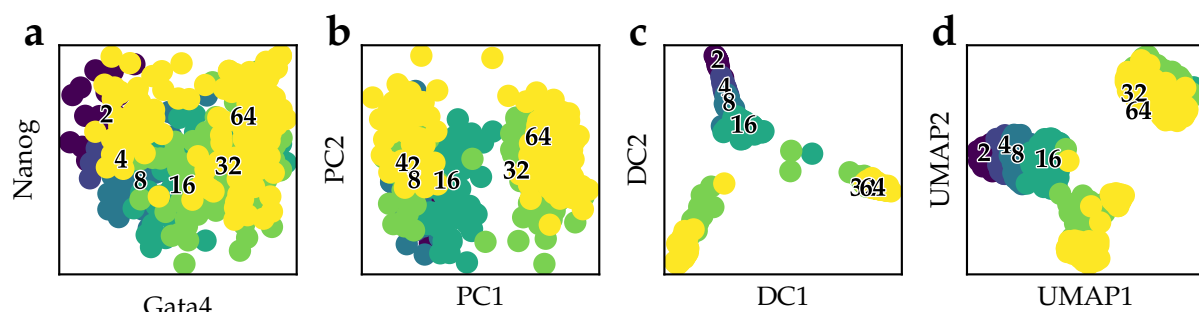


Figure 1.7 Four increasingly complex 2D embeddings of the same differentiating cells colored by age in days (Guo et al., 2010). **a)** Two carefully selected pluripotency markers are not able to separate the differentiation paths. **b)** A PCA is able to differentiate distinct clusters. **c)** Diffusion maps clearly show bifurcations and smooth transitions. **d)** UMAPs use machine learning to flatten multiple kinds of differences into a 2D plot, as in this case both bifurcation events present in the data.

Analysis of complex heterogeneity patterns requires methods of matching complexity to visualize their results. Complex developmental hierarchies and other relations in the data need to be faithfully represented. This commonly involves laying out points representing the cells in a flat plane while trying to maintain relationships such as developmental closeness (see Figure 1.7). **The challenge of finding a fitting low dimensional**

representation lies in the multiple scales biological processes take place in (Hasenauer et al., 2015) (see **Figure 1.8**). Therefore, as opposed to simple dimensional reduction methods used for distance calculations (see **Section 1.2.3**), the preferred methods for visualization take this multi-scale nature into account. Complex transforms are irreversible, so their resulting coordinates of cells in the visualization (the embedding) cannot be directly traced back to the input genes.



Figure 1.8 A so called hybrid image exemplifies how the same data can encode information on multiple scales. In this painting, the high frequency visual features are interpreted as a woman with a mirror while the low frequency details are interpreted as a skull (Oliva, Torralba, & Schyns, 2006).

(“All is Vanity” by Charles Allan Gilbert, 1892)

Without additional data providing a ground truth about subpopulation identity, or clearly separated clusters that are inherently assigned genes driving their embedding, the process of validating an embedding is not automatic, requiring creative reasoning and literature research. A panel of marker genes per cluster is derived from the data and cross-referenced with literature to annotate known cell types in the data. When discovering novel subpopulations, their adjacency to known ones can be used to make assumptions about their role and test them. An upcoming way to simplify the process are cell transcriptome atlases (Regev et al., 2017; Schaum et al., 2018) that can be used as references to automatically look up your cells and annotate them (Abdelaal et al., 2019).

An example for a relatively self contained scRNA-Seq analysis loop, this thesis presents DESTINY, a user-friendly, flexible and fast R tool for three closely related analysis steps (Angerer et al., 2016). It was created to enable dimensional reduction of both large, sparse scRNA-Seq data and qPCR data which benefits from explicit handling of missing values. The only previously existing tool was impractically slow and memory intensive when applied to scRNA-Seq data and did not support censored values present in qPCR data. Beyond routines for efficient computation and visualization of diffusion maps (see **Section 2.1.1.2**), DESTINY offers routines for diffusion pseudo time (Haghverdi et al., 2016), and gene relevance (see **Section 2.1.3**, Angerer et al., 2020). Its efficiency involves reuse of intermediate results from the diffusion map creation to calculate diffusion pseudo time between cells as well as to project new data into an existing diffusion map. Algorithms for analysis and visualization are at the forefront of discovery. They are however only part of the picture. A too narrow focus on individual steps or results also narrows the scope of what we consider doable. The bigger picture includes interoperability between tools. Progress is slow when one is bogged down by difficulties in menial tasks like file format conversion, tracking down ambiguously mapped gene names, and a faster machine to run an inefficient algorithm.

1.3 Organizing Analysis Workflows

In the scientific landscape of 2015 onwards, many tasks existed that scientists spent disproportionate time on rebuilding. By improving how individual steps integrate with each other, we could increase how often and well we interface with and build on each others' work.

Specifically growing data sizes (see **Figure 1.9**) and increasing data availability impact the field greatly (Angerer et al., 2017). Multiple theoretical possibilities mentioned in Angerer et al. (2017) are becoming reality. For example, cell atlases containing large amounts of multimodal data can now be used to infer labels for new data using transfer learning (Abdelaal et al., 2019).

In this section, I outline the ways in which programmatic frameworks improve reusability and therefore interoperability and reproducibility, specifically of scRNA-Seq analysis. I address how the field changed its approach to analysis workflows.

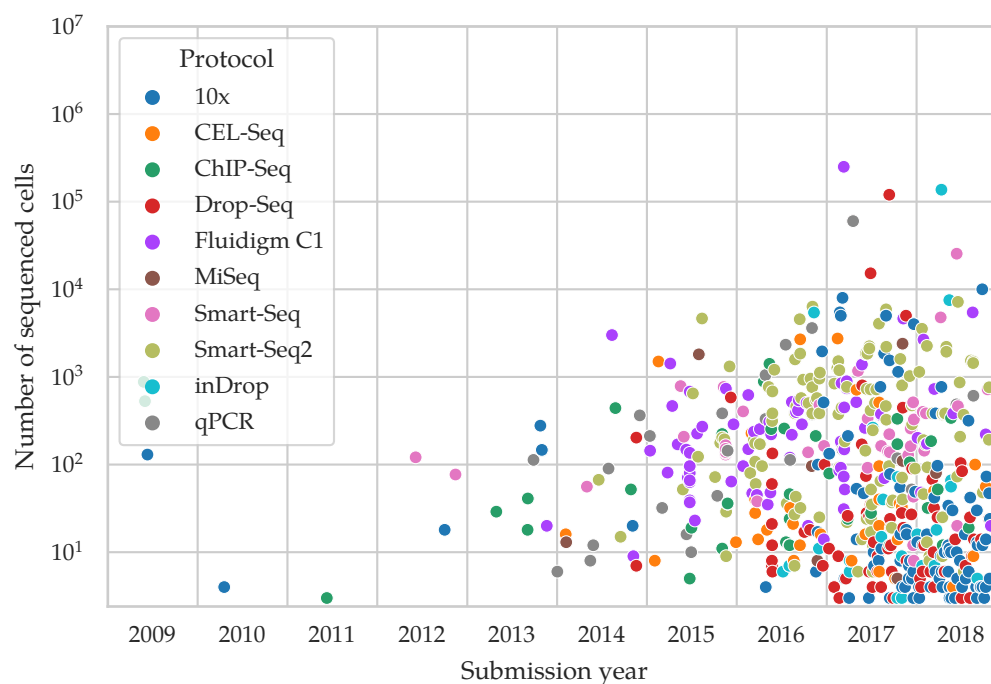


Figure 1.9 Single cell dataset sizes uploaded to Gene Expression Omnibus (GEO). This updated version of a figure from Angerer et al. (2017) shows the sustained trend of increasing datasets and cell numbers per dataset.

1.3.1 Analysis Pipelines: Status Quo

The steps necessary to reproduce the preprocessing and analysis of a publication are **bundled in an executable bundle** of scripts called a pipeline. Pipelines perform transformations on the data, **collect** statistics, and compile reports. Depending on the amount

of time and resources required to run the whole pipeline, it is desirable be able to only repeat part of the processing in case of an error. This is mostly achieved by regularly storing snapshot files representing the processing state between two subsequent sets of steps. Before my work in the area started, pipelines were mostly written in R or GNU MAKE and often executed a command line for each step, a procedure which by necessity consumed and produced a snapshot, but allows no choice when this high granularity is undesirable. Despite the field's increasing investment into machine learning algorithms implemented in Python, few tools were available in that language.

Communication with other languages or between pipeline steps involved reading **an (often textual) snapshot** from a file system, processing it, and writing it back to disk. This process involved long run times due to slow reading and writing, high storage space requirements and little resilience to errors. File corruption introduced e. g. via unreliable access to network storage could not be detected in text files, which lack checksums and other resilience precautions. In practice, the lack of frameworks **meant that** each tool or pipeline had to solve common problems by itself, such as hardening against the aforementioned reliability issues. This further wasted resources and increased the potential for errors. The resulting pipeline code published with papers was very application-specific and hard to generalize or share.

Even within analysis pipelines operating entirely in memory instead of on slow and unreliable files, individual tools were not necessarily designed to be compatible. Single tools could halt progress if they had insufficient performance for upcoming large datasets from microfluidic protocols. The tools' varying code quality, choice of default values and error reporting led to subtle incompatibility bugs. Furthermore, no data structures and file exchange formats suitable for scRNA-Seq existed, so workflows manually kept count matrices and annotation data frames in sync across filtering steps, increasing the potential for error. Furthermore, reproducibility was hard to guarantee in the presence of nondeterministic methods and a multitude of tool versions with subtly different defaults and behavior.

In the R world, the BIOCONDUCTOR project (Huber et al., 2015) provided tools for preprocessing and basic analysis, but little in the way of embeddings. DESTINY is well integrated into this ecosystem and, as mentioned earlier, closes functional gaps there.

No dimensionality reduction method except for the simplistic linear PCA provides a way to relate embedding location back to input variables. The popularity of nonlinear dimensionality reductions meant that there was a gap between using them as cell-level analysis tool and the differential expression (DE) analysis (Soneson & Robinson, 2018) as most popular gene-level analysis tool. DE requires discrete partitioning of the data, which is not well defined for data containing gradual transition of gene expression profiles between cells, and has the aforementioned issues with uncertainty propagation.

1.3.2 Frameworks for modular scRNA-Seq Analysis Pipelines

The original way to implement the aforementioned pipelines arose from the lowest common denominators of program interaction, summarized by Doug McIlroy as the UNIX philosophy (Salus, 1994):

Write programs that do one thing and do it well.

Write programs to work together.

Write programs to handle text streams, because that is a universal interface.

Text based file formats and pipeline steps realized as command line interfaces working on these files had their place for a long time. The approach results in automatic availability of intermediate steps as snapshots, in a human readable form, and to use basic text manipulation to work on them. Attempting to serve this lowest common denominator however lead to the aforementioned limitations when building pipelines.

The alternative is to use a framework, i.e. an opinionated set of data structures and routines that are tightly integrated but generic enough to be used as building blocks for pipelines or visualization tools. This tight integration allows to rely on information being tracked that allows the framework to provide necessary features, which would have to be rebuilt on a case-by-case basis without a framework. It also allows to bundle steps efficiently in memory, allowing control over when to invest the resources to store snapshots.

Due to their tighter coupling of data structures and algorithms, frameworks are specific to individual programming languages. For scRNA-Seq analysis, the major frameworks today are implemented in R and Python: SEURAT¹ (Butler et al., 2018; Satija et al., 2015), SCATER (McCarthy et al., 2017), and SCANPY (Wolf, Angerer, & Theis, 2018) are the most popular ones. Their data structures allow keeping gene and cell annotations and graphs in sync with the expression matrix while filtering and concatenating data (see **Section 2.2**). Interspersing steps from frameworks written in different languages stay possible via common serialization formats and in-memory conversion, while largely maintaining aforementioned advantages (Angerer, 2019b).

Reproducibility seems to be largely solved, as widely available container solutions such as DOCKER allow defining perfectly reproducible environments (Di Tommaso et al., 2015). Unlike standalone tools, the frameworks also consistently support random seeding for identical output of nondeterministic processes. The number of users of a framework or tool contributes to its quality and therefore the quality of studies relying on it, as more people report errors and special use cases, which increases the robustness of the frameworks towards those problems (Kim & Jiang, 2015). Bigger frameworks encompassing many tools have an easier time to test several permutations of their functionality than each tool would be able to when standing alone.

The R world is more fragmented than Python's: The BIOCONDUCTOR ecosystem provides annotated matrix data structures and tools for preprocessing scRNA-Seq data. SCATER provides extensive quality control and preprocessing but little for analysis

¹ Note that SEURAT in its original publication 2015 was no general purpose scRNA-Seq analysis framework yet, but a data alignment algorithm with additional helper routines.

(Luecken & Theis, 2019). SEURAT has opinionated preprocessing, analysis and plotting routines but is not integrated into BIOCONDUCTOR. Its data structure is SEURAT-specific and needs conversion to the more generic BIOCONDUCTOR data structures. Standalone tools in R like DESTINY (Angerer et al., 2016), SCMAP (Kiselev, Yiu, & Hemberg, 2018), BASiCS (Eling et al., 2018), or VELOCITY (La Manno et al., 2018) usually support BIOCONDUCTOR data structures or even just plain matrices and data frames.

In Python, many standalone tools **start supporting** the generic data structure ANN-DATA (Eraslan et al., 2019; Polański et al., 2020; Zappia, Phipson, & Oshlack, 2018). ANN-DATA supports reading and writing from and to several language agnostic interchange formats. It was the first annotated matrix data structure to support sparse expression matrices and therefore up to millions of cells. It provides its own format that supports disk backing for operations on data too large to be performed in memory, sparse or dense. ANN-DATA was developed alongside SCANPY in order to make the latter a highly modular framework (Wolf, Angerer, & Theis, 2018). SCANPY provides commonly used **building** blocks which, through sharing ANN-DATA as a mutual base, can be combined with specialized Python package for certain data types and algorithms such as scVELO (Bergen et al., 2019), SCIRPY (Sturm et al., 2020), MUON (Bredikhin, Kats, & Stegle, 2022), scVI-TOOLS (Gayoso et al., 2022), SQUIDPY (Palla et al., 2022), and more. SCANPY is now in widespread international use, continuously expanded with new algorithms, and together with **ANN-DATA core** of the **community governed** umbrella project SCVERSE (Zappia, Phipson, & Oshlack, 2018). Being based on NUMPY (Van Der Walt, Colbert, & Varoquaux, 2011), SCIPY (Virtanen et al., 2020), and other mature and highly performant Python technologies, SCANPY is very scalable. **Integrating well with JUPYTERLAB and JUPYTER notebooks (Kluyver et al., 2016), SCANPY lends itself to be used interactively to quickly iterate on analyses with close feedback from biologists. It also supports being used as a Python package to back visualization tools, web servers and traditional file-based processing pipelines.**

2 Methods

2.1 Dimensionality Reduction

Dimensionality reduction methods embed high dimensional data into a low dimensional space. They generally aim to reduce technical sources of variation (noise) while retaining variation produced by processes of interest. Dimensionality reduction methods can broadly be categorized by the resulting space's purpose, which is usually a 2D or 3D space for visualization or a latent space for further processing. Some methods also assign scores to each result dimension, which can e. g. be used to decide which dimensions to discard and which ones to keep for visualization or further processing. An ideal latent space embedding would exclusively capture all underlying processes' variability present in the data, and therefore match the intrinsic dimensionality of the data. A 2D or 3D embedding used as scatter plot coordinates serves as the basis of visualizations interpretable by humans. To this end, it can be colored or annotated using input variables and statistics to analyze static relations or dynamic processes of the data.

In the context of single cell RNA sequencing (scRNA-Seq) data, the high dimensional data are represented as a gene expression matrix x_{gc} with tens of thousands of genes $g \in \{1, \dots, G\}$, measured for up to approximately one million cells $c \in \{1, \dots, C\}$ at once (Zheng et al., 2017). Single cell expression matrices are very sparse since cells express a limited amount of genes at a time (Svensson, 2020). The intrinsic dimensionality of transcriptomic data is low when compared to this number of genes (Heimberg et al., 2016). Embeddings s_{pc} ($p \in \{1, \dots, P\}$) of most datasets therefore only require a comparatively low number of dimensions $50 \leq P \leq 100$ to capture all information necessary to reconstruct all known biological processes (Ding, Condon, & Shah, 2018; Lenz et al., 2016).

Dimensionality reduction methods can also be categorized into linear and nonlinear. Linear methods such as Principal Component Analysis (PCA, Abdi & Williams, 2010) use linear combinations of input dimensions as embedding dimensions, which means that distances in s directly relate to distances in x . Nonlinear methods such as t-Stochastic Neighborhood Embedding (t-SNE, Van Der Maaten & Hinton, 2008), Uniform Manifold Approximation and Projection (UMAP, McInnes, Healy, & Melville, 2018), or diffusion maps (Coifman et al., 2005) attempt to maintain locality in the data. This means that local neighborhoods of samples in x are still close neighbors in s (red circles in **Figure 2.1b**). Some nonlinear dimensionality reduction methods also maintain global distances to a degree (UMAP and diffusion maps, not t-SNE).

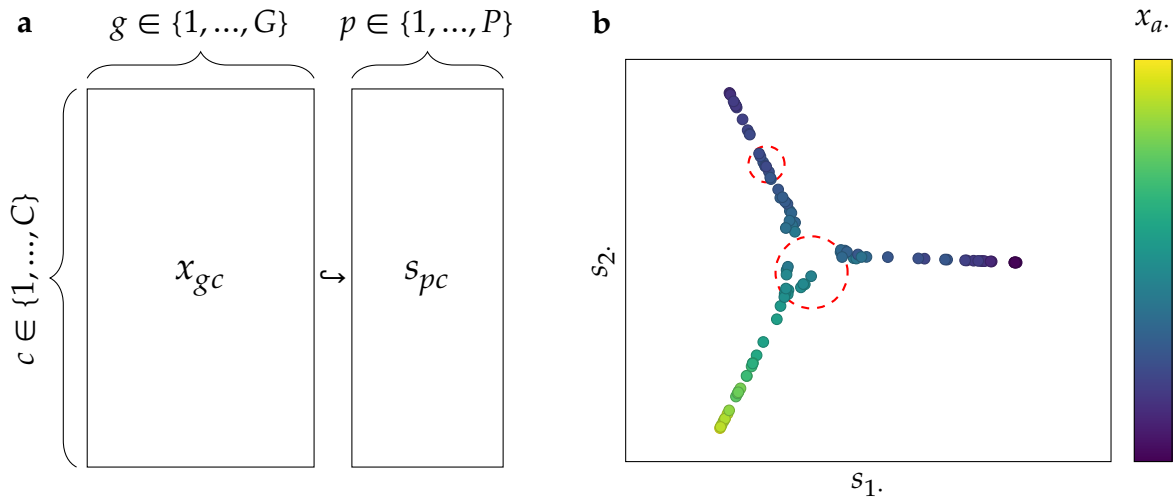


Figure 2.1 Reducing dimensions using global and local distances. **a)** Dimensionality reduction methods embed samples (such as cells) from a high dimensional space (such as gene expressions) to a low dimensional space ($x_{gc} \hookrightarrow s_{pc}$, $G \gg P$). They can be used in data preprocessing or as coordinates in visualizations. **b)** A scatter plot using embedding coordinates visualizes similarities between cells' gene expression profiles x_a and serves as a reference frame for visualizing further information. It can be annotated to explain processes and label cell groups or colored with any kind of cell label to see it in context. Here, the embedding shows a bifurcation of the cells and a gradual change of gene a 's expression x_a along the branches. If we know that gene a is e. g. a marker for early development, this indicates that cells in the bottom left branch are young and undifferentiated, and the two other branches are differentiated.

2.1.1 Spectral Decomposition

Spectral decomposition is the act of transforming data into components that each capture a different source of variance in the data. These components are scored by their ability to explain the data's structure. By keeping components with high enough scores as embedding dimensions, spectral decompositions can be used as dimensionality reduction methods. This is most useful as a very fast preprocessing step to define a less noisy latent space reflecting the inherent dimensionality of the data, but components can also be directly visualized (Linderman et al., 2019).

2.1.1.1 Principal Component Analysis

PCA is a simple spectral decomposition (Abdi & Williams, 2010). Principal components (PCs) p are weighted products of input variables (such as gene transcripts g), scored by the variance each PC explains. A PCA embedding s_{pc} can be defined in multiple ways,

one of which is based on the singular value decomposition of a centered data matrix Z derived from the raw data x_{gc} :

$$Z = (z_{gc}) = \left(x_{gc} - \frac{1}{C} \sum_{c=1}^C x_{gc} \right) \quad (2.1)$$

Singular value decomposition is a factorization of the matrix $Z = U\Sigma V^T$, where $U^T U = I_g$, $V^T V = I_c$, and Σ is a $g \times c$ diagonal matrix with all entries $\in \mathbb{N}_0$. The embedding s_{pc} can be defined in terms of the left singular vectors U and the singular value matrix Σ :

$$(s_{pc}) = S = (U\Sigma)^T \quad (2.2)$$

As each PC ideally represents one source of variation, data with up to around four linearly separable sources of variation can realistically be interpreted using all PC combinations as coordinates for a series of 2D scatter plots. While the intrinsic dimensionality of transcriptomic data is low, the number of PCs necessary to faithfully represent all biological processes is around 50-100 (Lenz et al., 2016). Picking a number of PCs in this range, for example using the jackstraw algorithm (Chung & Storey, 2015) is a popular preprocessing step for downstream analysis.

There are a number of related methods that can serve a similar purpose, such as independent component analysis (Bell & Sejnowski, 1995) or factor analysis (Stephenson, 1935). Furthermore, single cell transcriptomics specific extensions exist, such as the application of GPLVM in Buettner et al. (2014), and the PCA extensions ZINB-WaVE (Risso et al., 2018) and GLM-PCA (Townes et al., 2019), which account for high numbers of zeros and over-dispersion of scRNA-Seq data.

2.1.1.2 Diffusion Maps

Diffusion maps provide another spectral decomposition of data. Their practicality for data analysis was first shown by Coifman et al. (2005).

A diffusion map s_{pc} has as components p the eigenvectors of a matrix of transition probabilities T_{cd} between cells c and d . The transition probabilities are derived from a diffusion model applied to the cells' expression profiles x_{gc} . There are several flavors of diffusion models, parametrized over the chosen distance function *dist* and the kernel width σ . Using a common σ for all cells can lead to T_{cd} becoming disjunct, rendering its Eigen decomposition undefined. This led to the development of locally scaled diffusion maps, in which the kernel width σ_c is locally inferred for each cell. A locally scaled Gaussian function is:

$$Y_c(c') = \sqrt[4]{\frac{2}{\pi\sigma_c^2}} \exp\left(-\frac{\|c' - c\|^2}{\sigma_c^2}\right) \quad (2.3)$$

Where σ_c is an estimate of the local density of x_{gc} around c using c 's k th-nearest neighbor $NN_k(c)$:

$$\widehat{\sigma}_c = \text{dist}(c, \text{NN}_k(c)) \quad (2.4)$$

Since Gaussian functions are wave functions, the kernel $K(c, d)$ is defined by the interference of two wave functions centered at c and d :

$$\begin{aligned} K(c, d) &= \int_{-\infty}^{\infty} Y_c(c') Y_d(c') dc' \\ &= \sqrt{\frac{2\sigma_c\sigma_d}{\sigma_c^2 + \sigma_d^2}} \exp\left(-\frac{\|c - d\|^2}{2(\sigma_c^2 + \sigma_d^2)}\right) \end{aligned} \quad (2.5)$$

With a kernel function defined, the connectivities W_{cd} between cells are defined as follows:

$$W_{cd} = \frac{K(c, d)}{Z(c)Z(d)} \quad (2.6)$$

$$Z(c) = \sum_{d \in \{1, \dots, C\}} K(c, d) \quad (2.7)$$

More commonly today, W is approximated. One approach is to only define W_{cd} for $d \in \{\text{NN}_k(c)\}$ instead of all $d \in \{1, \dots, C\}$, another is employed by UMAP (see **2.1.2**), which uses fuzzy simplicial sets derived from σ_c instead of discrete sets of nearest neighbors.

In any case, a transition probability matrix T can be obtained by row-normalizing W :

$$\tilde{Z}(c) = \sum_{d \in \{1, \dots, C\}} W_{cd} \quad (2.8)$$

There are different possible definitions for T_{cd} . The symmetrical version has several computational advantages, including performance and the ability to reuse the sparsity structure of W :

$$T_{cd} = \sqrt{\tilde{Z}(c)}^{-1} W_{cd} \sqrt{\tilde{Z}(d)}^{-1} \quad (2.9)$$

Compared to machine learning methods like t-SNE that embed data nondeterministically (see **2.1.2**), diffusion maps preserve global structure and the pseudo-temporal ordering of cells better (Haghverdi, Buettner, & Theis, 2015). The first diffusion component is often sufficient to directly capture and visualize bifurcation events in differentiating cells.

2.1.2 Learned Embeddings

Machine learning based dimensionality reduction methods use optimization to either learn an embedding directly, or a function mapping high-dimensional data to embedding coordinates. A common optimization target is to represent high-dimensional relationships in the data as faithfully as possible in a 2D or 3D embedding that can be displayed on a screen or paper.

The machine learning driven method t-SNE remains the most popular dimensionality reduction (Kobak & Berens, 2019). T-SNE optimizes an embedding so that similarities in the high-dimensional space x_{gc} are maintained in the low-dimensional embedding s_{pc} . Similarly to locally scaled diffusion maps, it uses distributions with kernel widths based on x_{gc} 's local densities to calculate similarities between cells. This process exaggerates short distances and de-emphasizing long distances, resulting in t-SNE's sensitivity towards its perplexity parameter, which scales local density estimates and therefore kernel widths. Similarity between cells is denoted as $p_{j|i}$, which stands for x_j 's probability density in a distribution centered at x_i . T-SNE uses a Gaussian distribution to calculate similarities $p_{j|i}$ within the high dimensional data. The iterative optimization process starts with a randomly initialized embedding s_{pc}^0 . For each iteration k , it performs the following steps: It calculates the similarities $q_{j|i}^k$ within the embedding s_{pc}^k using a t-distribution (Student, 1908). Then it performs gradient descent to modify the embedding in a way that the new embedding s_{pc}^{k+1} 's similarities $q_{j|i}^{k+1}$ are closer to the similarities $p_{j|i}$ of the high dimensional data x_{gc} . It stops iterating if the similarity matrices converge sufficiently or a maximum number of iterations is reached.

The more recent method UMAP works similarly, while claiming to represent global structures in the data more faithfully (Becht et al., 2018). Its more continuous and global structure also makes it a viable dimensionality reduction for preprocessing similarly to the aforementioned spectral decomposition methods (see 2.1.1). Kobak and Linderman (2019) however showed that t-SNE and UMAP perform very similarly with the same data-driven initialization and selection of meta-parameters. Single cell transcriptomics specific methods exist in this space as well, such as scLVM (Buettner et al., 2015), a successor of the earlier GPLVM (Buettner & Theis, 2012). Their performance criteria however don't lend themselves well to the heterogeneity analysis of much more than few hundreds of cells performed today. This led to other approaches, such as learned embeddings modeled explicitly to perform scRNA-Seq preprocessing, e. g. single cell variational inference (scVI, Lopez et al., 2018). Alternatively, one can preprocess the data in a way that improves embeddings by general purpose methods, e. g. by denoising data using a deep count autoencoder (DCA, Eraslan et al., 2019).

A disadvantages of machine learning based methods is their nondeterministic nature: due to the non-convex nature of the optimization, its converging to local optima can result in unstable inference of certain cells' neighborhoods in the embedding (Hinton & Roweis, 2003). Another problem is that machine learning methods that learn an embedding directly instead of an embedding function are unable to project new data into an existing embedding. There are approaches to remedy this for UMAP and t-SNE (Van Der Maaten, 2009).

2.1.3 Gene Relevance

Linear dimensionality reduction methods like PCA are rarely used for visualizing scRNA-Seq data anymore. Nonlinear methods are preferred as their flexibility allows them visualize more information at once in 2D or 3D plots. They however lack the linear

methods' intrinsic relation from reduced dimensions to input genes. Such knowledge about which genes drive an embedding allows to identify target genes for subsequent analysis.

The gene relevance method was developed in order to derive this relation from scRNA-Seq data and any given embedding (Angerer et al., 2020). It requires no previous cluster detection algorithm and is able to find genes that cannot be identified using differential expression analysis. The method defines a gene relevance score R_{gc} for gene g and cell c , as well as summary scores called global gene relevance $GR_r(g)$ and local gene relevance $LR_r(g, \Psi)$ for any given set of cells Ψ . It also defines a gene relevance map as a scatter plot using the embedding coordinates and coloring each cell by its label l_c , which denotes the gene driving c 's embedding location most. The gene relevance score R_{gc} is a measure of the extent in which a gene g drives the embedding coordinates s_{pc} of a specific cell c . It can be derived from the gradient ∇f_{gc} of the function f mapping a embedding s_{pc} to the gene expression x_{gc} :

$$\nabla f_{gc} = \left(\frac{\partial x_{gc}}{\partial s_{1c}}, \dots, \frac{\partial x_{gc}}{\partial s_{pc}} \right) \quad (2.10)$$

This gradient can be estimated using local differences $\frac{\partial x}{\partial s} \approx \frac{\Delta x}{\Delta s}$ within each cell c 's kNN neighborhood $NN_k(c) \wedge n \neq c$. To address the high number of zeroes in scRNA-Seq data, the gradient is only defined for $x_{gc} \neq 0$:

$$\widehat{\nabla f_{gc}} = \begin{pmatrix} NA, & \text{if } x_{gc} = 0 \\ \text{median}_{n \in NN_k(c) \wedge n \neq c} \frac{x_{gn} - x_{gc}}{s_{pn} - s_{pc}}, & \text{otherwise} \end{pmatrix} \quad (2.11)$$

The gene relevance score R_{gc} is defined as the euclidean norm of this P -dimensional gradient vector:

$$R_{gc} = \|\nabla f_{gc}\|_2 = \sqrt{\sum_{p=1}^P (\nabla f_{gc})_p^2} \quad (2.12)$$

Local gene relevance $LR_r(g, \Psi)$ is a measure of how much a gene g is driving the embedding given a set of cells $\Psi \subseteq \{1, \dots, C\}$. Local gene relevance depends on a rank cutoff $r \in \{2, \dots, |\Psi|\}$.

$$LR_r(g, \Psi) = \frac{1}{|\Psi|} \sum_{c \in \Psi \wedge x_{gc} \neq 0} [\text{rank}(R_{gc}) < r] \quad (2.13)$$

With the Iverson (1962) bracket notation:

$$[P] = \begin{cases} 1, & \text{if } P \text{ is true} \\ 0, & \text{otherwise} \end{cases}, \text{ where } P \text{ is any predicate.} \quad (2.14)$$

The set of cells of interest Ψ can be obtained by metadata such as experimental markers or the labels from a clustering algorithm. Another alternative is the visualization of local relevance by dividing the embedding into shaped bins, which can be colored according to their respective local relevance.

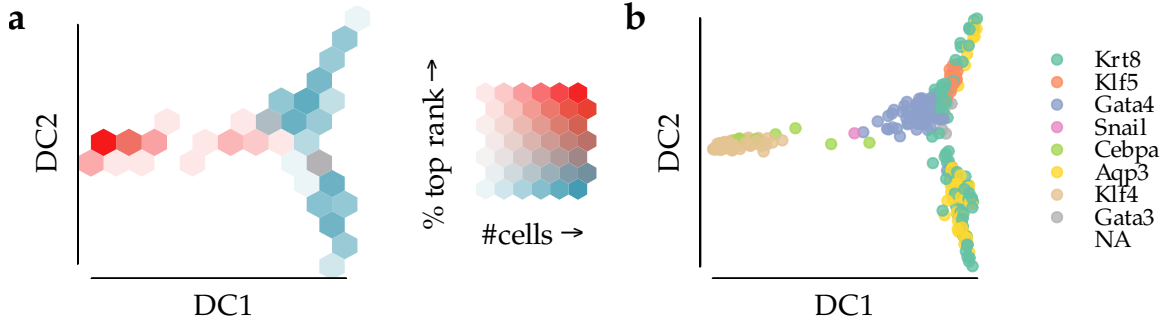


Figure 2.2 Gene relevance visualizations on a diffusion map embedding allow to reason about the role individual genes play for the embedding. Our example data from Guo et al. (2010) contains differentiating cells, with their differentiation state controlled by transcription factors like Klf4. We therefore expect those same genes to have a strong influence on the location of each cell in an diffusion map embedding derived from the data. **a)** Local gene relevance of the gene Klf4 and local density for each hexagonal bin. This score can be calculated for any given gene in order to investigate how much it drives each cell's location in the embedding. As expected, a transcription factor like Klf4 has strong effect on positioning cells in the left arm of the embedding. **b)** Gene relevance map. Calculating all scores allows to label each cell by the gene which drives its location in the embedding most. Klf4 shows up next to Cebpa as the strongest driver for cells in the left arm.

Global gene relevance $GR_r(g)$ is a score that emphasizes genes driving the embedding for large parts of the data. It is defined as the local relevance of genes $\{1, \dots, C\}$:

$$GR_r(g) = LR_r(g, \{1, \dots, C\}) \quad (2.15)$$

Using it as a global gene score yields a similar result as PCA loadings.

Gene relevance maps were developed since global relevance cannot be used to detect genes driving the embedding of small subpopulations, and local relevance requires defining discrete sets of cells to score. They label each cell with the gene having the highest gene relevance score for that cell. Because this labeling is subject to noise, m smoothing steps over the scores of each cells' neighborhood are employed to obtain less and more stable labels l_c^m .

$$l_c^m = \underset{g \in \Omega \wedge x_{gc} \neq 0}{\operatorname{argmax}} \begin{cases} R_{gc}, & \text{if } m = 0 \\ \frac{1}{k} \sum_{n \in \text{NN}_k(c)} [l_n^{m-1} = g], & \text{otherwise} \end{cases} \quad (2.16)$$

Gene relevance is robust towards its rank cutoff parameter r and reasonably robust towards its smoothing parameter m (Angerer et al., 2020), given that strong smoothing $m \geq 3$ often results in all cells sharing the single most globally relevant gene as label. The implementation as part of the R package `DESTINY` (Angerer et al., 2016) is sufficiently efficient in terms of computational time and memory so that quick iterations are possible. Like other parts of `destiny DESTINY`, it allows control over the distance metric used, and handles missing values and outliers in the data.

2.2 ANNDATA

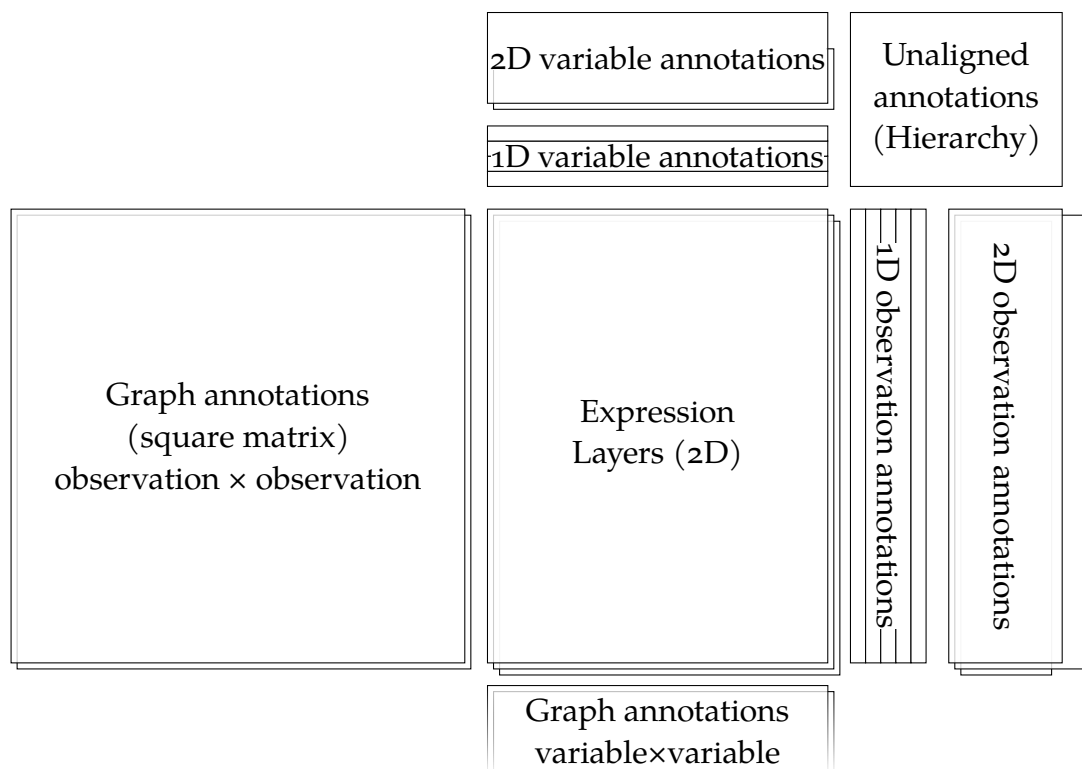


Figure 2.3 ANNDATA is structured around a central stack of variable $\times$ observation matrices (layers) and a hierarchy of unstructured annotations. Each dimension, variables and observations, can be annotated with vectors (1D), matrices (2D) and graphs (2D, square matrices).

ANNDATA is a general purpose data structure for annotated two dimensional data (see **Figure 2.3**). This is in contrast to tidy data, where each variable type has its own column: in our case that would mean one column each for gene ID, cell ID, expression value, and each annotation (Wickham, 2014). Tidy data is useful for small datasets, but redundant and therefore memory-expensive for data with large amounts of variables. It would also be computationally wasteful to use the tidy format for data is commonly analyzed using matrix operations on a observation $\times$ main-variable matrix.

`AnnData`'s structure therefore matches the annotated observations-by-variables ($c \times g$) matrix format most biological snapshot data is analyzed in.² This includes single cell imaging, where this matrix contains each segmented cell c 's average fluorescent channel g value. In RNA-Seq or quantitative real-time polymerase chain reaction (qRT-PCR, Nurmi et al., 2002), g stands for a gene's expression in a tissue sample or a single cell. Each observation c or variable g can be annotated with different labels used in any step of an analysis pipeline. Annotations that exist from the start can stem from technical sources such as experimental batches or different data sources such as images from flow cytometry (Fulwyler, 1965). They can also be derived from knowledge about a variable's or observation's origin like the genome a gene exists in or the tissue sample a cell came from. Other annotations can be added by preprocessing and analysis steps such as cell classification (Li et al., 2017).

`AnnData` can best be understood as its set of attributes. Each aligned attribute denoted with `[]` is Python mapping of names to aligned arrays. `AnnData` maintains this alignment, keeping matrices and annotations in sync when filtering and concatenating `AnnData` objects.

`.obs_names` and `.var_names` are arrays of unique names or IDs for for each `AnnData` dimension. Names for observations c are stored in `.obs_names`, and variable g names in `.var_names`.

`.layers[]` is the central element of each `AnnData` object, a mapping of names to 2D matrix layers, each holding all variables g measured for all observations c . Missing values in an observation can be represented by the special pseudo-value `NA`. Each layer has one homogeneous scalar type able to represent it in sufficient IEEE 754 floating-point precision or integer size (IEEE, 2019). The layers can be dense `NumPy` arrays (Van Der Walt, Colbert, & Varoquaux, 2011) for normalized or densely measured data, as well as `SciPy` sparse matrices (Virtanen et al., 2020) for scRNA-Seq data with many zeros.

`.X` is an accessor for the layer that is used by default in preprocessing or analysis operations.

`.obs[]` and `.var[]` are `PANDAS` data frames (Mckinney, 2010). Their columns hold 1D annotations such as cluster labels, alternative gene IDs besides `.var_names`, or ranking indices.

`.obsm[]` and `.varm[]` hold 2D annotations, such as experimentally determined cell coordinates or dimensionality reduction coordinates.

² In the R world, R's native data frame matches this orientation, while count matrices are its transpose: $g \times c$

- `.obsp[]` and `.varp[]` hold weighted graphs, represented as sparse square matrices. An example is neighborhood graphs containing cell-cell distances calculated with a specific distance metric.
- `.uns[]` holds unstructured hierarchical annotations, such as analysis algorithm parameters. They are the only kind of annotation not modified when filtering observations or variables.

ANNDATA supports several file formats it can read and write to for long term result storage, pipeline checkpoints, file exchange, and compatibility with existing experimental protocols and other tools and pipelines. This includes its own `.h5ad` format based on HDF5 (Folk et al., 2011), LOOM ([<linnarson2017>](#)), UMI-TOOLS (Smith, Heger, & Sudbery, 2017), MATRIX MARKET (Boisvert & Remington, 1996), as well as Microsoft Excel and textual spreadsheets. ANNDATA supports disk-backed operations in its native format, allowing operations on data too big to hold in memory. For quick integration of ANNDATA with BIOCONDUCTOR packages (Huber et al., 2015) or SEURAT (Butler et al., 2018; Satija et al., 2015), ANNDATA can be converted to and from SINGLECELLEXPERIMENT (Lun, Risso, & Korthauer, 2017) using ANNDATA2RI (Angerer, 2019b). Due to ANNDATA's generic structure, and its conversion and serialization options, it serves as a backbone for analysis pipelines. SCANPY and other tools can interact by adding and reading annotations, as well as reading, modifying and adding layers.

2.3 SCANPY

SCANPY is a Python package providing functions and classes for scRNA-Seq preprocessing, analysis, and visualization (see [Section 1.2.3](#), [1.2.4](#), and [1.2.5](#), respectively). Scanpy is popular and widely used, see [Section 3.3](#) for details. Its main design philosophy is ease of use without compromising performance or interaction with other tools. This allows analyzing of up to a million cells in reasonable time (Wolf, Angerer, & Theis, 2018). SCANPY's ease of interaction with other tools was ensured by developing and designing it in parallel with ANNDATA and avoiding a SCANPY-specific data structure. SCANPY also integrates well with the popular JUPYTERLAB, allowing to iterate on an analysis in an interactive environment that preserves code, plots, and output as a report file called "notebook" (Kluyver et al., 2016). Combined with disk-backed operations or saving of snapshots, this allows to interrupt a workflow at any point or move it to a different machine. A basic SCANPY workflow utilizes all aforementioned features of ANNDATA, establishing naming conventions by which subsequent steps interact. For example, storing batch labels as `.obs['batch']` will make sure that SCANPY's batch correction algorithms use them by default. SCANPY's dimensionality reduction methods store their coordinates as `.obsm['X_<method>']`, while SCANPY provides an embedding plotting function with a parameter to choose the coordinate basis, as well as a plotting function for each method with sensible defaults. SCANPY reads data either directly from

an experimental protocol format like 10x GENOMICS' CHROMIUM (Zheng et al., 2017) or a generic interchange format supported by ANNData's read functions.

2.3.1 Preprocessing

It allows to preprocess data using a variety of quality control models and visualizations. Those help to filter low quality cells and genes, and to transform raw transcript counts into expression profiles matching the expectations of subsequently applied analysis tools. Filters are based on count and mitochondrial gene content (Illicic et al., 2016; Islam et al., 2014), while transforms reduce batch effect and library size biases (Luecken & Theis, 2019). The variability score of each gene (Satija et al., 2015; Zheng et al., 2017) can be stored as annotation and either used as further, rigorous filtering or by specific processing steps. Otherwise, preprocessing generally modifies the default layer `.X` and removes genes and cells from layers and aligned annotations.

2.3.2 Analysis

After finalized preprocessing, SCANPY analyses generally start by pre-computing commonly used information. This includes a PCA with usually 50-100 components to be used as basis for distance calculations within the data and from reference data to new data. For highly performant access of distances within the data, a nearest neighbor graph is calculated. The graph serves as the basis of several different nonlinear, locality based analyses. This includes certain dimensionality reduction methods like UMAP (McInnes, Healy, & Melville, 2018), graph layout (Islam et al., 2011; Weinreb et al., 2018) and diffusion maps, as well as clustering algorithms such as LEIDEN (Traag, Waltman, & van Eck J., 2019), diffusion pseudo time (Haghverdi et al., 2016) and PAGA (Wolf et al., 2019). SCANPY also offers data integration similar to Polański et al. (2020), and several gene scores. These allow analyzing either the influence of given known gene groups such as cell cycle genes (Satija et al., 2015) and the differential expression between clusters in one of different flavors such as the variance adjusted t-test or Wilcoxon rank-sum test (Soneson & Robinson, 2018; Wilcoxon, 1945).

2.3.3 Visualization

SCANPY has rich visualizations based on Matplotlib (Hunter, 2007) for each step from basic quality control to complex analysis. Their flexibility is rooted in them being generic over the annotation column used for coloring or statistic summarization (see **Figure 2.4**). The final step of SCANPY analyses consists of converting the complete JUPYTER notebook, select visualizations, and the final data to publishable formats using few simple commands (Wolf, Angerer, & Theis, 2018).

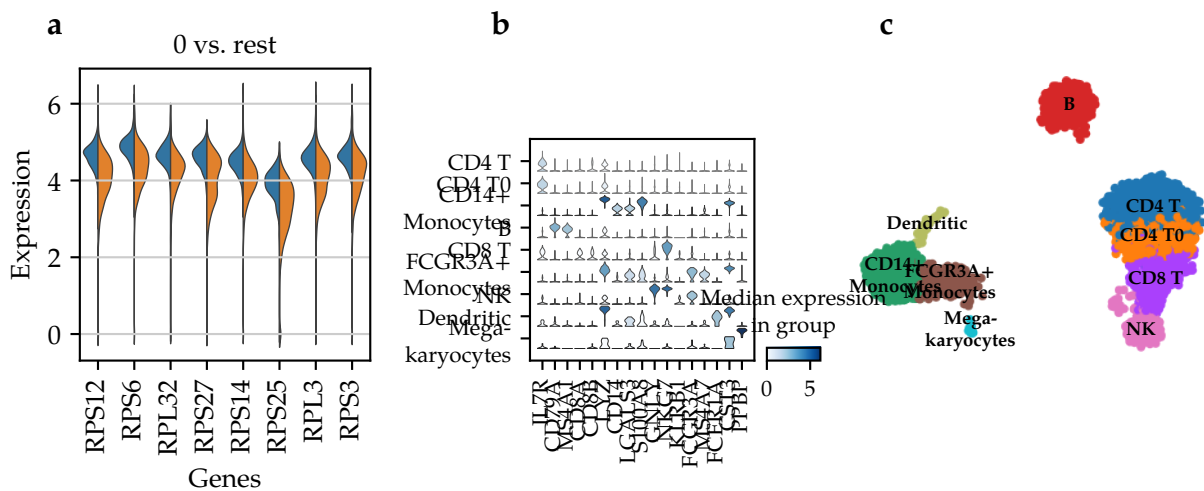


Figure 2.4 SCANPY flexibly computes annotations from scRNA-Seq data and visualizes results combined with previous knowledge. A example cluster analysis starts by determining cell clusters with the LEIDEN algorithm: `sc.tl.leiden(adata)`

a) Marker genes for clusters are determined with a Wilcoxon rank-sum test.

```
sc.tl.rank_genes_groups(adata, 'leiden', method='wilcoxon')
```

Violin plots show how well each gene's expression profile distinguishes the cluster.

```
sc.pl.rank_genes_groups_violin(adata, groups='0', n_genes=8)
```

b) The clusters are renamed according to literature research of their marker genes.

```
adata.rename_categories('leiden', cluster_names)
```

Violin plots show the clusters' expression profiles for each marker gene.

```
sc.pl.umap(adata, color='leiden', legend_loc='on data')
```

c) An UMAP embedding colored by the manually labelled LEIDEN-detected clusters.

```
sc.pl.dotplot(adata, marker_genes, groupby='leiden')
```

2.4 BART-Seq (demultiplexing and quantification)

Compared to classical plate-based RNA sequencing protocols (see **Figure 1.4**), BART-Seq is targeted. It trades the ability to measure all available transcripts in an unbiased way for the ability to inexpensive, highly sensitive measurement of a limited set of target transcripts for thousands of samples (Uzbas et al., 2019). High sensitivity is maintained

when using single cells as samples. BART-Seq consists of three steps; design, experiment, and quantification.

The design step involves algorithms to design primer pairs matching amplicons – such as genes – and barcodes identifying samples. The primer design algorithm optimizes primer pairs to stably dimerize with their target amplicons. Each pair contains two primers binding to the 5' ends of each strand of the target amplicon, forward and reverse. The set of primer pairs is split up into one set of forward and one of reverse primers. The barcode design algorithm minimizes the chance of misassigning barcodes through sequencing errors by maximizing their mutual alignment distances. The set of barcodes is split into two halves arbitrarily, one half per primer set. The results are the sequences for a forward and reverse set of barcodes, a forward and reverse linker, and a forward and reverse primer per amplicon.

For the experimental step, all barcode-linker and reverted complement (r. c.) primer-linker sequences are synthesized in sufficient quantities. Separately for each barcode, all combinations of forward primers and forward barcodes are created by annealing barcode-linker and r. c. primer-linker strands arbitrarily, filling in the resulting incomplete double strands, and removing the r. c. strands. Analogously all possible reverse barcode-linker-primer strands are synthesized per reverse barcode. Each well of a sequencing plate receives a different combination of forward and reverse barcode-linker-primer assemblies, allowing to address $(\frac{1}{2}\#\text{barcodes})^2$ wells. The experiment concludes like a regular paired-end sequencing run (Fullwood et al., 2009), resulting in a raw read library containing strands of barcode-linker-read.

In the quantification step, the barcodes, linkers and low-quality bases are cut off the raw read sequence. The shortened read sequences are then tagged with their corresponding designed barcode if possible. This process assigns barcodes with up to one mismatching base using the algorithm of Aho and Corasick (1975), as long as the assignment is still unique – reads that could be tagged with one of multiple barcodes that are one mismatch away from each other are discarded. The tagged reads are then aligned to the targeted amplicons using HISAT2 (Kim et al., 2019). Amplicons are counted per barcode combination according to one of two strategies. The conservative strategy only counts where both reads aligned to the same amplicon, while the sensitive strategy also counts when one of both is unaligned. Neither strategy counts one of the aligned amplicons when there is a mismatch. The count of overall full matches, half matches, and mismatches is included in the various statistics collected about the demultiplexing and quantification process.

3 Summary of contributed articles

This chapter contains summaries of the articles I made substantial contributions to during the course of my work as PhD candidate.

In general, I have contributed to a class of changes that scientists are adopting in their approach to scRNA-Seq analysis. My contributions focused on improving the ability of scientists to apply the theoretical steps outlined in **Section 1.2** to real data in praxis. I worked on every computational step of the scRNA-Seq analysis process described in **Section 1.2** except for alignment, with a specific focus on steps involving a count matrix.

My contributions to the field of scRNA-Seq data analysis include research, development, and extension of algorithms, data structures, pipelines, frameworks, and APIs.

3.1 destiny: diffusion maps for large-scale single-cell data in R (Angerer et al., 2016)

Summary: In this article, we described the initial scope of `DESTINY`, an R package implementing diffusion maps. Diffusion maps were chosen for their at the time unique ability to capture smooth nonlinear transitions in single cell transcription data and its robustness towards noise. `DESTINY` was the first user friendly and practical implementation of the algorithm, suited for both interactive use by data scientists and use in other packages. Its practicality is a result of high performance making it scale from personal computers to high performance machines, as well as algorithmic enhancements adapting the algorithm to scRNA-Seq and qPCR data.

`DESTINY`'s scalability and computational efficiency is mainly achieved by reusing the sparsity structure of the sparse distance matrix to perform element-wise operations. To this end, `DESTINY` symmetrizes its kNN matrix and uses the symmetric variant of diffusion maps. Other adaptations for efficiency include such initializing based on a k-nearest-neighbor search, full sparse matrix support for input data, as well as taking advantage of an approximate Eigen decomposition algorithm. This efficient design has been adopted by `SCANPY` for its many tools that use a kNN search as preprocessing step.

`DESTINY` also efficiently reuses intermediate results such as the transition matrix for its other functions. Those include diffusion pseudo time (DPT, Haghverdi et al., 2016) and the ability to project a second set of data into an existing diffusion map. Apart from the sparsity support making it suitable for scRNA-Seq, explicit support for missing and censored values also make `DESTINY` fit to support qPCR data. The R package is continuously maintained and has been adapted to make use of current best practices such as an adaptable diffusion map kernel, plotting with `GGPLOT2` and storing expression data in the `SINGLECELLEXPERIMENT` data structure.

This integration and flexibility makes `DESTINY` especially user friendly. It is also easy to use without detail knowledge by being able to run without specifying parameters. To

this end, `DESTINY` locally scaled diffusion maps, and heuristically derives a fitting number of nearest neighbors. This nonparametric diffusion map variant is also used by `SCANPY` today. When opting for a global scale instead, `DESTINY` will derive a suitable kernel width by default.

`DESTINY` contains dedicated plotting routines for diffusion maps and DPT that directly integrate into the popular plotting package `GGPLOT2`. `Destiny` started providing perceptually continuous color-maps before they have been included in the major plotting packages of Python and R. Without them, accurate assessment of any kind of color-mapped visualizations cannot be guaranteed (Borkin et al., 2011; Moreland, 2017). `DESTINY` supports `BIOCONDUCTOR`'s annotated expression matrix classes as well as plain matrices and data frames.

`DESTINY` can utilize a number of distance metrics including euclidean, cosine, and rank correlation distance. Unlike other available R tools, `DESTINY` uses the chosen distance metric also for kNN search, which has been shown to improve results (Hu et al., 2016). `DESTINY` can be used fully deterministically by setting a random seed. `DESTINY`'s API is flexible enough to be used as an analysis tool in its own right, as well as as a diffusion map backend for integrated analysis pipelines like `SCATER` (McCarthy et al., 2017). It also served as a flexible base for the implementation of DPT and gene relevance (see below).

`Destiny` has been cited 175 times at the time of writing and has for example been used for dimensional reduction and pseudo-time inference in analyses. This includes hematopoietic stem and progenitor cell differentiation (Nestorowa et al., 2016), an immune atlas of clear cell renal cell carcinoma (Chevrier et al., 2017), development in the human adult brain (Lake et al., 2018) and an atlas of the aging drosophila brain (Davie et al., 2018).

My contribution: For the tool, I started with a basic diffusion map implementation from Haghverdi, Buettner, and Theis (2015). It already supported missing and censored values, while I developed all scRNA-Seq related extensions and implemented the first published version by myself. I implemented all further enhancements of the tool. I conceived all optimizations improving its memory and time efficiency and therefore enabling its use on a personal computer, where less efficient tools needed to be ran on a powerful server. `DESTINY` initially supported raw standard sparse matrix data structures and later gained support for `SINGLECELLEXPERIMENT`. As a consequence, `DESTINY` became one of the first scRNA-Seq analysis tools able to handle sparse matrices and therefore millions of cells on a personal computer. I helped Laleh Haghverdi in developing the nonparametric diffusion map variant during the parallel development of `DESTINY` and DPT, paving the way for `SCANPY`'s use of the same. I chose to default to perceptually continuous color-maps, which put `DESTINY` ahead of its time in guaranteeing accurate assessment of information encoded into plot colors. I contributed to a few of `DESTINY`'s dependencies to support random seeds for determinism. For the publication, I wrote the first draft, which I then completed together with other contributors. I designed the main figure to which Maren Büttner contributed cell type annotations, and created all supplementary figures.

3.2 Single cells make big data: New challenges and opportunities in transcriptomics (Angerer et al., 2017)

Summary: In this review article, we provide a selective view on the history of single cell data, focusing on the growing number of captured cells per transcriptomics experiment. We evaluate the reasons for capturing more cells, the qualitative changes in the captured data, and the consequences of these changes in terms of a shift in where challenges lie. Capturing a high number of cells provided the necessary resolution to identify certain rare subpopulations, as well as for accurate inference of developmental trajectories. As sequencing depth decreases just as cell numbers increase, data size per experiment does not change much. However the tools to handle the data change, as e.g. expression matrices need to be stored in a sparse format which those tools need to support, and the noise per datum increases. Challenges can be approached with a variety of attack angles, that usually range from emulation of older shapes, such as imputing noisy sparse data to make it smooth and dense, to approaches modelling the new properties. The challenges can change rapidly: while smaller scRNA-Seq require regularization to handle their intrinsic “small n large p” properties, newer experiments capturing 10s of thousands of cells already no longer suffer from this problem.

My contribution: Starting from an idea by Fabian Theis, I performed the data analysis supporting the publication’s primary thesis, showing conclusively the growth of scRNA-Seq data in both volume per experiment and number of experiments. I designed and produced the figures with feedback from other contributors. ote the major part of the publication, with parts contributed by other contributors.

3.3 Scanpy: large-scale single-cell gene expression data analysis (Wolf, Angerer, & Theis, 2018)

Summary: In this article, we introduced SCANPY, a framework for single cell analysis and visualization. SCANPY is a framework for scRNA-Seq preprocessing, analysis, and visualization in Python. AnnData is a data structure for annotated observation data including scRNA-Seq counts and expressions. It is designed to be generic and independent of SCANPY.

SCANPY is designed around AnnData, storing the results of each processing step in an expression matrix layer or an aligned annotation (see **Figure 2.3**). AnnData can be stored on disk in various formats for snapshots, data interchange, and interoperability with command line tools. Its structure allows serialization in the file-tree like formats HDF5 (Folk et al., 2011) and ZARR (<miles2022>). They, unlike Python’s “pickle” format are not bound to individual implementation versions and therefore allow for

reliable long term storage. For data too large to hold in memory, `AnnData` supports disk-backed operations on its HDF5-based `.h5ad` files. For details about the implementation of `AnnData` and `SCANPY`, see **Sections 2.2** and **2.3**

`AnnData`'s and `SCANPY`'s modular and flexible design philosophies allows `SCANPY` routines to be effortlessly interspersed with preprocessing and analysis steps of other tools, including R tools via `SINGLECELL EXPERIMENT` (Angerer, 2019b; Lun, Risso, & Korthauer, 2017). `AnnData` has been used for this purpose by several scRNA-Seq analysis tools since the publication of `SCANPY`, such as `scVI` (Lopez et al., 2018), `DCA` (Eraslan et al., 2019), `DESC` (Li et al., 2019), or `BBKNN` (Polański et al., 2020). Its `.h5ad` format is supported by visual analysis pipelines like `GALAXY` (Afgan et al., 2018) or `CELLXGENE` (Megill et al., 2020). `SCANPY` is widely used and popular in the scRNA-Seq analysis community (Zappia, Phipson, & Oshlack, 2018). It was named Genome Biology's milestone article of the year 2018 (Genome Biology, 2020). It is used for large projects like the `HUMAN CELL ATLAS` (Regev et al., 2017), `TABULA MURIS` (Schaum et al., 2018), or `EMBL-EBI's` expression atlas, which uses `SCANPY` for downstream analysis of all single cell experiments (Papatheodorou et al., 2020). The `HUMAN CELL ATLAS` and `COVID19 CELL ATLAS` (Wilk et al., 2020) chose `AnnData`'s `.h5ad` format as their data interchange format. At the time of writing, `SCANPY` is cited by over 300 studies, review papers and tools.

My contribution: For the Python packages, I designed `AnnData` and the foundational API structure of `SCANPY`. With F. Alexander Wolf I implemented the initial version of both packages, where he focused on performance while I focused on API and data structure design. I contributed to the implementation of some algorithms included in `SCANPY`. I designed `AnnData`'s on-disk format together with F. Alexander Wolf, driving the choice of HDF5 as a reliable format, which led to aforementioned adoption by large projects and popular ools. I contributed to the manuscript after F. Alexander Wolf wrote the first draft.

3.4 BART-Seq: cost-effective massively parallelized targeted sequencing for genomics, transcriptomics, and single-cell analysis (Uzbas et al., 2019)

Summary: In this article, we developed BART-Seq, a targeted RNA-Seq protocol usable at the single cell or bulk levels. See Uzbas et al. (2019) and **Section 2.4** for BART-Seq, a transcriptomics protocol that allows to quantify many samples at once. It is set apart from untargeted single cell and bulk RNA-Seq protocols due to its high sensitivity and low cost. BART-Seq can use single cells or groups of multiple cells as samples. Each of its samples is tagged by combining paired-end reads with pairs of computationally designed barcodes. It is therefore possible distinguish high numbers of samples while sensitively measuring the specified target amplicons, such as expressed genes. An advantage of the predictable barcoding over randomized barcodes is that sample identity is maintained, allowing

for arbitrarily much information attached to each sample of a single batch, such as treatment labels. BART-Seq has an unique demultiplexing and counting strategy that can be tuned for sensitivity or conservativity (see **Section 2.4**). The counting strategies and visualization routines for the protocol are available as a Python package allowing fine grained programmatic access (Angerer, 2019a). Part of this package is the BART-Seq equivalent to CELL RANGER or zUMIs: a SNAKEMAKE pipeline (Köster & Rahmann, 2012) that performs all steps of the protocol to create count matrices, statistics and visualizations.

My contribution: I designed the demultiplexing and counting strategy together with Fatma Uzbaz and Micha Drukker. I implemented the Python package, and built the pipeline wrapping the package for use with minimal configuration. For the publication, I wrote the section about demultiplexing and contributed to the remaining sections, mainly to the primer design section. I demultiplexed the final version of all data used in the publication using my pipeline.

3.5 Automatic identification of relevant genes from low-dimensional embeddings of single-cell RNA-seq data (Angerer et al., 2020)

Summary: In this article, we present gene relevance, a novel method for inference of the contributions individual variables have on the location of data in an embedding. This algorithm is the latest addition to the DESTINY package, building upon its efficient routines. The method was inspired by PCA loadings, a intrinsic property of the linear dimensionality reduction method PCA. These loadings are the linear factors scaling a variable to define its role in a principal component. Using PCA as dimensional reduction method for expression data therefore allows to see each gene's contribution to a cell's position in a plot. Gene relevance generalizes the idea to make it applicable to nonlinear dimension reduction methods. It is designed to identify genes that drive the local positioning of a cell in an embedding by calculating the differential of each gene-cell combination. This involves estimating the partial derivatives of the gene expression with respect to the embedding location by looking at cell neighborhoods. This allows to detect locally relevant genes without a discrete clustering step, and therefore without data reuse as a result of using inferred cluster labels as input for gene level analysis. The article shows robust derivation of both obvious candidates such as hemoglobin genes for hematopoietic development, as well as interesting but less obvious genes. We also present an easy to use interface for calculating gene lists and plotting as part of the DESTINY package.

My contribution: The initial idea comes from Fabian Theis, while I designed the changes making it applicable to real data with A. Scialdone and C. Marr. I implemented the algorithm as an extension of DESTINY's data structures and API, offering a rich and user friendly way to interact with for all gene relevance concepts, including plotting

routines. I also extended the idea to include scores for defined groups of cells (such as clusters) or the whole data for a global relevance score per gene. The method is described in detail in **Section 2.1.3**. For the publication, I designed and performed its data analysis, which Antonio Scialdone interpreted. I wrote the first draft, while David Fischer and Carsten Marr contributed to the mathematical description. I completed writing the manuscript with feedback from Carsten Marr.

4 Conclusion and Outlook

Within few years, single cell RNA-Seq has grown to an extensive and active field of science, with new trends and opportunities. The largest current challenges are the ever-growing dataset sizes, and the integration of different kinds of data (Angerer et al., 2017; Nature Methods, 2020). Going up to hundreds of thousands of cells, the data allows us to abandon the notion of discrete cell types and to explore the continuous processes at play. Destiny (Angerer et al., 2016) and the gene relevance method presented in this thesis prove to be valuable contributions in this direction.

As demands on reliability at scale grow along demands posed to algorithms, more practices from software engineering are finding their way into biological science. For example, extensible and efficient data format standards are born from the collaboration of scientists to supplant the ad-hoc text-based formats we used to incompatibly modify to suit situational needs. Open source analysis frameworks have been designed to offer an open and welcoming environment for new as well as established analysts and method developers. My contributions towards more reproducible, scalable and interconnectible scientific programming have helped these changes along.

4.1 Multiresolution scRNA-Seq Analysis

The analysis of continuous processes driving cellular heterogeneity has been enabled by fluorescent tagging and single cell imaging. Their targeted nature as well as their monetary and time costs however renders them unsuitable for unbiased heterogeneity analysis. While careful selection of parameters like sequencing depth is vital, scRNA-Seq is an unbiased and inexpensive method able to detect rare cells (Torre et al., 2018). This makes it uniquely suited for heterogeneity analyses. A rarely addressed property of cellular heterogeneity is its multiresolution nature: multiple processes at different scales and locality shape heterogeneity of either subpopulations or all cells.

Analysis algorithms in this area have evolved alongside the protocols able to sequence hundreds of thousands of cells. Suitable for the low sample sizes of bulk RNA-Seq, simple linear dimension reduction methods like PCA were used to visualize early single cell studies. Nonlinear methods like diffusion maps, t-SNE or later UMAP have been developed and continue being improved to fit the nature of transcriptome data. But while they capture both continuity and heterogeneity at multiple resolutions, their output does not contain directly interpretable information about processes at different resolutions. Methods like the growing multitude of trajectory inference algorithms or gene relevance complement them within the same constraint (Saelens et al., 2019). Of all methods known to the author, only PAGA so far distinctively dissects continuous multiresolution heterogeneity (Wolf et al., 2019). To analyze this foundational attribute of cellular processes, a multiresolution variant of gene relevance would be of great use.

4.2 Multimodal SCANPY and AnnData

Recent developments in the joint measurement of scRNA-Seq data and multiple other data types (modes) at single cell resolution have enabled multimodal analysis. With it giving ripe opportunity for analysis development, Nature Methods has named single cell multimodal omics its method of the year 2019 (Nature Methods, 2020). Efremova and Teichmann (2020) provide a review of the different single cell modes that became jointly measurable with scRNA-Seq in short succession: RNA velocity, spatial location, surface protein abundances, epigenetic modifications, and chromatin accessibility.

If there is no 1:1 relation between samples, transfer learning can be used to migrate labels from one dataset or mode to another (Abdelaal et al., 2019). This technique has been successfully employed to link data in and to the HUMAN CELL ATLAS (Regev et al., 2017). SCANPY, AnnData and projects based on the two have partial support for multimodal data, using the existing capabilities of AnnData.

RNA Velocity RNA Velocity is a model to infer trajectories of cell differentiation from the combination of unspliced pre-mRNA with spliced mRNA (La Manno et al., 2018). Based on SCANPY and AnnData, scVELO is an user friendly implementation and extension of this model (Bergen et al., 2019). The tool uses which uses AnnData's `.layers` to distinguish between spliced and unspliced transcripts.

Spatial location Measuring spatial location of cells together with transcriptomic heterogeneity allows to analyze spatial trends such as the location of markers in embryonic development. The success of the 10X GENOMICS CHROMIUM platform indicates that the upcoming VISIUM platform will make spatial information ubiquitous in the near future (Maniatis et al., 2019). Novel approaches like in-situ sequencing also allow capturing 3D spatial coordinates via STARMAP (Wang et al., 2018). Spatial analysis in SCANPY is straightforward, as 2D or 3D space can be stored in AnnData's `.obsm`. The current development of spatial conventions and plotting routines will encourage the development of SCANPY based spatial algorithms similar to the R tools trendsceek (Edsgård, Johnsson, & Sandberg, 2018) and SpatialDE (Svensson, Teichmann, & Stegle, 2018).

Surface protein abundances Measuring surface protein abundances is difficult but necessary in cases where transcriptomics and proteomics are poorly correlated, such as with posttranslationally modified proteins. Measuring those abundances together with scRNA-Seq data has become possible through Cellular Indexing of Transcriptomes and Epitopes by sequencing (CITE-seq, Gayoso et al., 2019).

Epigenetics Two epigenetic modes can be combined with scRNA-Seq data (Henriksson, 2019). Chromatin accessibility in single cells has been possible to measure for some years using single cell Assay for Transposase-Accessible Chromatin using Sequencing (scATAC-Seq, Buenrostro et al., 2013). DNA methylation can be measured now using the recently developed single cell Cleavage Under Targets and

Tagmentation (scCUT&Tag, Kaya-Okur et al., 2019). For analysis of scRNA-Seq data combined with both modes, *EPISCANPY* offers a *SCANPY*-integrated workflow (Danese et al., 2019).

AnnData and *SCANPY* currently lack true support for multimodal data where different modes can be stored for one set of observations. This enhancement would allow to store annotations for each mode separately while allowing joint filtering of observations. A further feature would be to store connections between features of different modes in a graph, such as which transcript (scRNA-Seq) encodes which surface protein (CITE-Seq). The current version of *AnnData* substantially increased usability and therefore analysis efficiency for exclusively scRNA-Seq data. Therefore a multimodal *AnnData* should do the same for the analysis of data like scCUT&Tag or CITE-Seq, which contain a second high-dimensional feature mode in addition to transcriptomic data. Existing workflows and Python packages that use the current version of *AnnData*, like *EPISCANPY*, will benefit from this upcoming multimodal extension of *AnnData*.

The combination of different data types within a single experiment enabled unprecedented insight into disease. Extending *AnnData* and *SCANPY* for joint analysis of spatial location and scRNA-Seq data promises to bring their ergonomic benefits and performance to histologic patient data analysis.

4.3 New Dimensions of *SCANPY* Scalability

SCANPY has proven to be scalable in terms of CPU and memory through its efficient implementation, multiprocessing capabilities and *AnnData*'s disk-backed mode. For a successful open source project, more is necessary than just performance and ease of use. Open source projects need to be scalable in terms of maintenance structure and trust (de Laat B., 2010).

The next steps of the *SCANPY* project in this regard will focus on community building. To foster openness and inclusivity, *SCANPY* will adopt a code of conduct (CoC). CoCs have been shown to increase diversity in a project, while higher diversity is linked to increased productivity (Tourani, Adams, & Serebrenik, 2017). To uncouple progress with the available amount of free time of the de-facto decision makers of the project, a steering committee will be formed. Its membership will be open to long-term contributors, and its meeting notes publicly available. Automations for quality and performance criteria, such as *SCANPY*'s high performance GPU-driven parts, similarly streamline the contribution process and will be extended. This will make friendly and helpful information available regardless of the available free time of experienced contributors. Additionally, inclusion criteria of external tools to the *scanpy* website and the `scanpy.external` module will be formalized and made transparent. Communication channels with the main projects of the R world are already established and will continue to be used to ensure interoperability.

These changes should be able to encourage more junior developers and data scientists to start contributing to the project. More active contributors in turn means that key

knowledge of the project is shared by more people, which reduces reliance on the continued effort of a small group of individuals.

Further improvements can come from the professional development world, especially if some contributors can invest time regularly. A bigger group of regular contributors can result in adopting agile principles. This would result in a more structured pace of development, with regularly scheduled releases and the ability to plan which features are being focused on.

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